

Generative AI for transition metal chemistry (making the novel, exceptional?)



David Balcells

III CAMLC Workshop

June 2-5, 2026 | Zaragoza, Spain

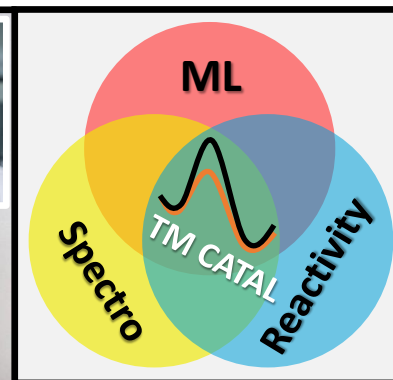
Acknowledgements: People and Funding



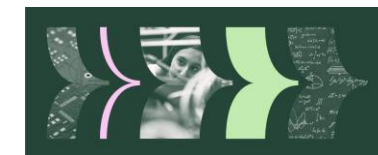
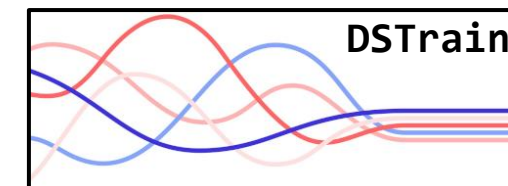
UiO Computational Catalysis Group

Alejandra, Nisha, Lucía, Arron, Hannes, Trond, Magnus

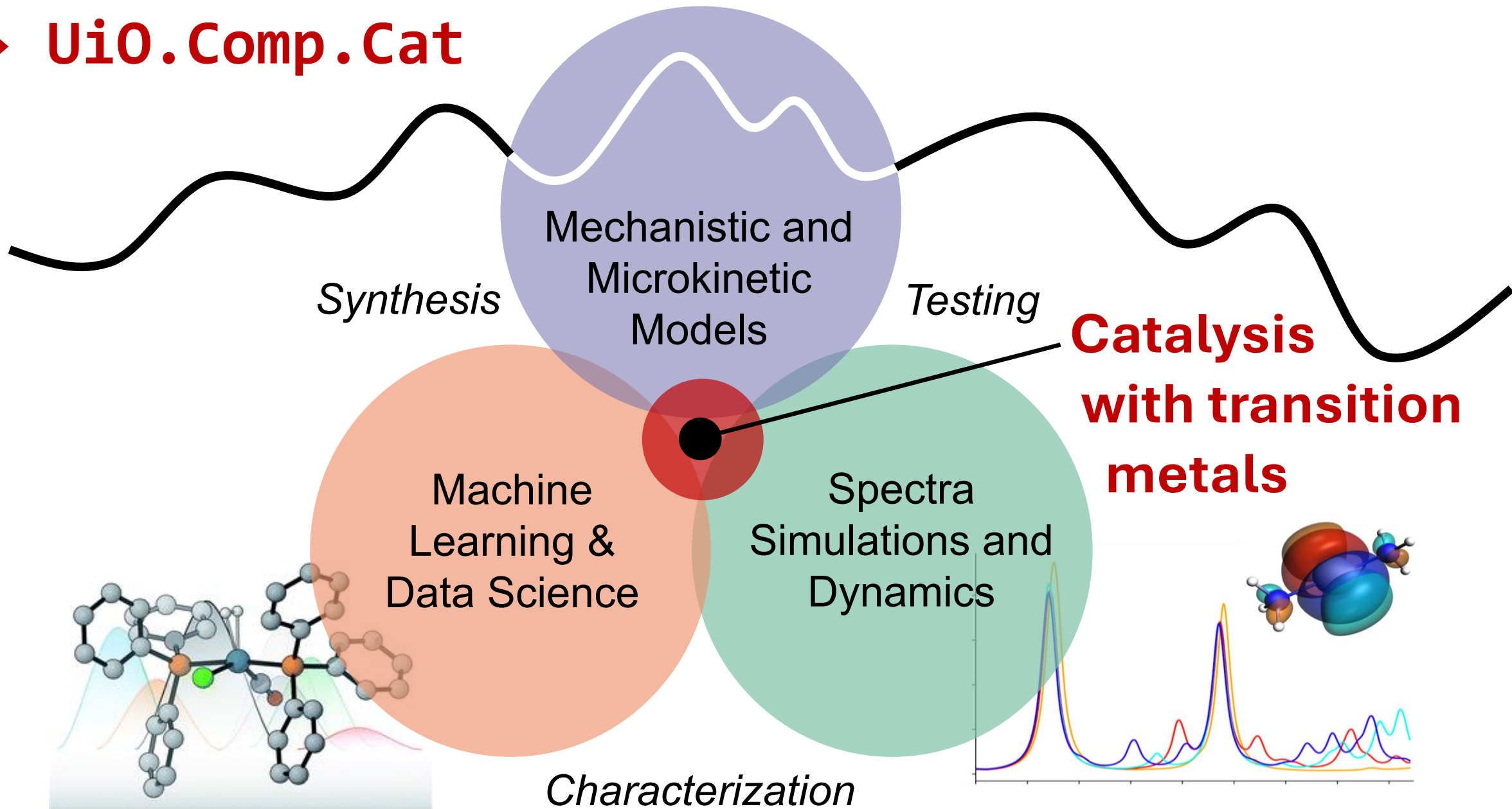
Ainara



Hylleraas SFF Center
Department of Chemistry
University of Oslo

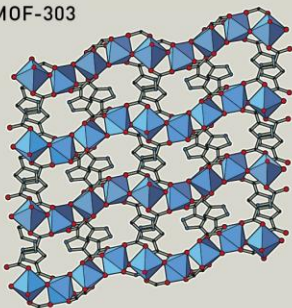


> UiO.Comp.Cat



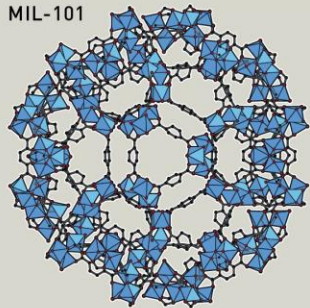
UiO-67 MOF highlighted by the Nobel Prize in Chemistry 2025

MOF-303



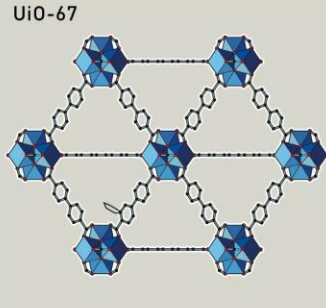
MOF-303 can capture water vapour from desert air during the night. When the sun heats up the material in the morning, potable water is released.

MIL-101



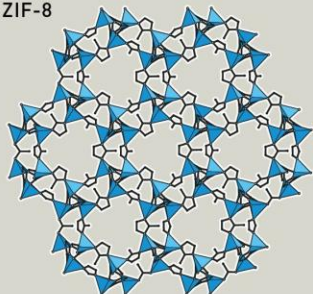
MIL-101 has gigantic cavities. It has been used to catalyse the decomposition of crude oil and antibiotics in polluted water. It can also be used to store large quantities of hydrogen or carbon dioxide.

UiO-67



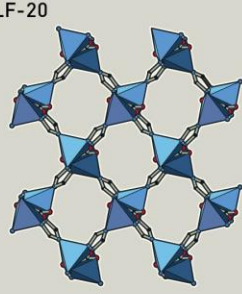
UiO-67 can absorb PFAS from water, which makes it a promising material for water treatment and the removal of pollutants.

ZIF-8



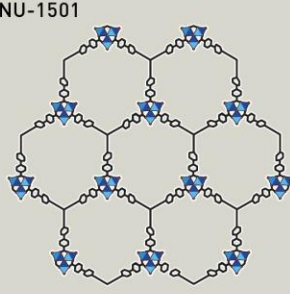
ZIF-8 has been used experimentally for mining rare-earth elements from wastewater.

CALF-20



CALF-20 has an exceptional capacity to absorb carbon dioxide. It is being tested in a factory in Canada.

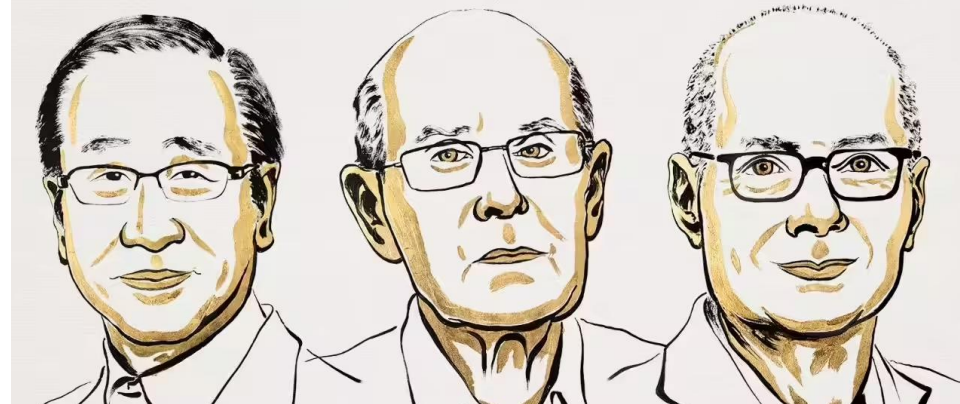
NU-1501



NU-1501 has been optimised to store and release hydrogen at normal pressure. Hydrogen can be used to fuel vehicles, but in ordinary high-pressure tanks the gas is extremely explosive.



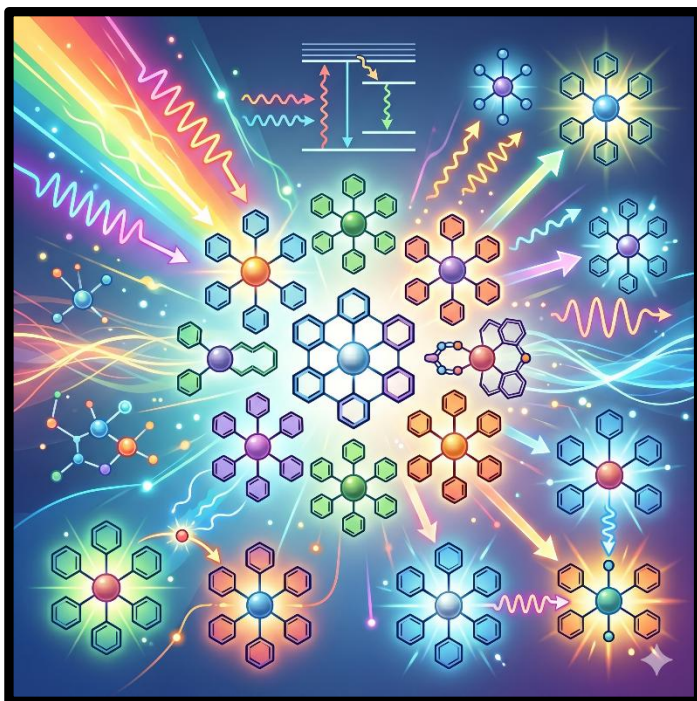
THE NOBEL PRIZE
IN CHEMISTRY 2025



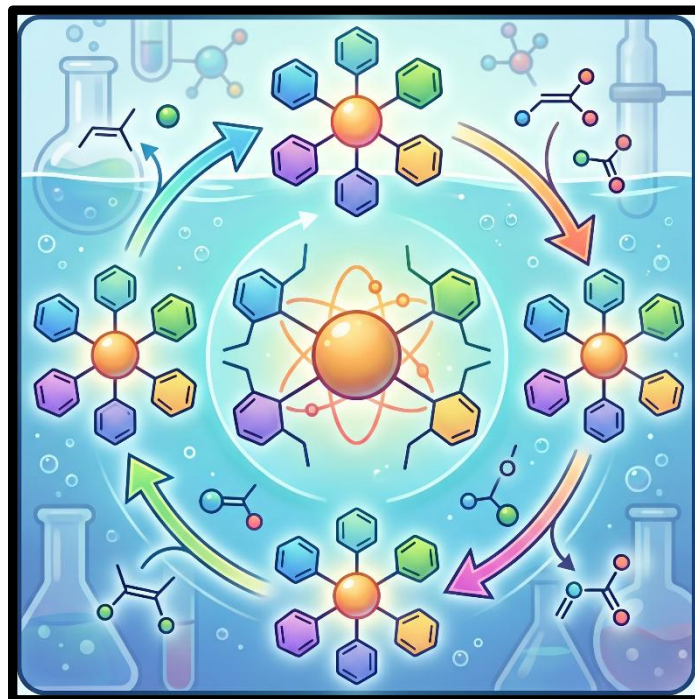
UiO = Universitet i Oslo

Stable Zr MOFs with
carboxylate linkers

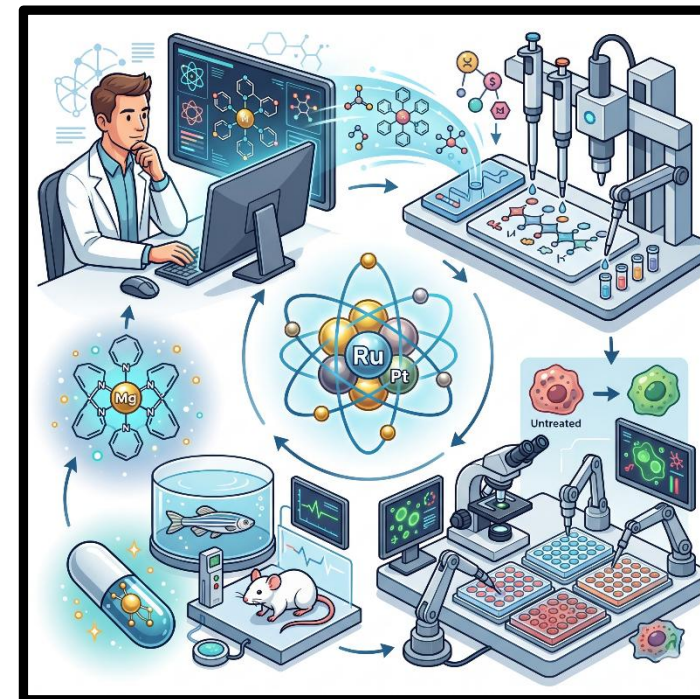
Why are we interested in the chemistry of the transition metals?



**MOLECULAR
PHOTONICS**



**HOMOGENEOUS
CATALYSIS**



**METALLODRUG
DISCOVERY**

AI is a powerful tool to explore the chemical space

Visible Universe: 10^{24} stars, 10^{80} atoms

Chemical space: $\mathcal{W} > 10^{200}$ compounds

Weininger number (no rigorous derivation)

Periodic Table of the Elements																																																																																									
1 H Hydrogen 1.008																	2 He Helium 4.003																																																																								
3 Li Lithium 6.941	4 Be Beryllium 9.012									5 B Boron 10.81	6 C Carbon 12.011	7 N Nitrogen 14.007	8 O Oxygen 15.999	9 F Fluorine 18.998	10 Ne Neon 20.180																																																																										
11 Na Sodium 22.990	12 Mg Magnesium 24.305	13 Al Aluminum 26.982	14 Si Silicon 28.086	15 P Phosphorus 30.974	16 S Sulfur 32.06	17 Cl Chlorine 35.45	18 Ar Argon 39.948																																																																																		
19 K Potassium 39.098	20 Ca Calcium 40.078	21 Sc Scandium 44.956	22 Ti Titanium 47.88	23 V Vanadium 50.942	24 Cr Chromium 51.996	25 Mn Manganese 54.938	26 Fe Iron 55.845	27 Co Cobalt 58.933	28 Ni Nickel 58.693	29 Cu Copper 63.546	30 Zn Zinc 65.38	31 Ga Gallium 69.723	32 Ge Germanium 72.63	33 As Arsenic 74.922	34 Se Selenium 78.96	35 Br Bromine 79.904	36 Kr Krypton 83.798																																																																								
37 Rb Rubidium 85.468	38 Sr Strontium 87.62	39 Y Yttrium 88.906	40 Zr Zirconium 91.224	41 Nb Niobium 92.906	42 Mo Molybdenum 95.94	43 Tc Technetium 98.906	44 Ru Ruthenium 101.07	45 Rh Rhodium 102.905	46 Pd Palladium 106.42	47 Ag Silver 107.868	48 Cd Cadmium 112.415	49 In Indium 114.818	50 Sn Tin 118.710	51 Sb Antimony 121.757	52 Te Tellurium 127.6	53 I Iodine 126.905	54 Xe Xenon 131.29																																																																								
55 Cs Cesium 132.905	56 Ba Barium 137.327	57-71 Lanthanides		72 Hf Hafnium 178.49	73 Ta Tantalum 180.948	74 W Tungsten 183.84	75 Re Rhenium 186.207	76 Os Osmium 190.23	77 Ir Iridium 192.222	78 Pt Platinum 195.084	79 Au Gold 196.967	80 Hg Mercury 200.59	81 Tl Thallium 204.38	82 Pb Lead 207.2	83 Bi Bismuth 208.98	84 Po Polonium 209	85 At Astatine 210	86 Rn Radon 222																																																																							
87 Fr Francium 223	88-103 Actinides		104 Rf Rutherfordium 261	105 Db Dubnium 262	106 Sg Seaborgium 266	107 Bh Bohrium 264	108 Hs Hassium 277	109 Mt Meitnerium 268	110 Ds Darmstadtium 271	111 Rg Roentgenium 272	112 Cn Copernicium 285	113 Nh Nihonium 284	114 Fl Flerovium 289	115 Uu Ununpentium 288	116 Lv Livermorium 293	117 Uus Ununseptium 294	118 Uuo Ununoctium 294																																																																								
Lanthanides <table><tr><td>57</td><td>La</td><td>58</td><td>Ce</td><td>59</td><td>Pr</td><td>60</td><td>Nd</td><td>61</td><td>Pm</td><td>62</td><td>Sm</td><td>63</td><td>Eu</td><td>64</td><td>Gd</td><td>65</td><td>Tb</td></tr><tr><td>138.905</td><td>140.12</td><td>140.908</td><td>144.24</td><td>144.913</td><td>144.913</td><td>150.36</td><td>150.36</td><td>157.25</td><td>158.925</td><td>162.50</td><td>164.930</td><td>167.26</td><td>171.947</td><td>174.967</td><td>175.937</td><td>187.945</td><td>188.905</td></tr></table> Actinides <table><tr><td>89</td><td>Ac</td><td>90</td><td>Th</td><td>91</td><td>Pa</td><td>92</td><td>U</td><td>93</td><td>Np</td><td>94</td><td>Pu</td><td>95</td><td>Am</td><td>96</td><td>Cm</td><td>97</td><td>Bk</td></tr><tr><td>227.033</td><td>232.038</td><td>231.036</td><td>238.029</td><td>237.048</td><td>244.041</td><td>244.064</td><td>244.064</td><td>247.071</td><td>251.077</td><td>252.083</td><td>257.10</td><td>261.108</td><td>265.108</td><td>269.109</td><td>273.109</td><td>287.103</td><td>289.103</td></tr></table> Alkali Metal Alkaline Earth Transition Metal Radio Metal Transition Nonmetal Semimetal Nonmetal Gas Lanthanide Actinide																		57	La	58	Ce	59	Pr	60	Nd	61	Pm	62	Sm	63	Eu	64	Gd	65	Tb	138.905	140.12	140.908	144.24	144.913	144.913	150.36	150.36	157.25	158.925	162.50	164.930	167.26	171.947	174.967	175.937	187.945	188.905	89	Ac	90	Th	91	Pa	92	U	93	Np	94	Pu	95	Am	96	Cm	97	Bk	227.033	232.038	231.036	238.029	237.048	244.041	244.064	244.064	247.071	251.077	252.083	257.10	261.108	265.108	269.109	273.109	287.103	289.103
57	La	58	Ce	59	Pr	60	Nd	61	Pm	62	Sm	63	Eu	64	Gd	65	Tb																																																																								
138.905	140.12	140.908	144.24	144.913	144.913	150.36	150.36	157.25	158.925	162.50	164.930	167.26	171.947	174.967	175.937	187.945	188.905																																																																								
89	Ac	90	Th	91	Pa	92	U	93	Np	94	Pu	95	Am	96	Cm	97	Bk																																																																								
227.033	232.038	231.036	238.029	237.048	244.041	244.064	244.064	247.071	251.077	252.083	257.10	261.108	265.108	269.109	273.109	287.103	289.103																																																																								

J. Chem. Inf. Model. 2024 special issue
on “Chemical Compound Space Exploration”

1st Chemical Compound Space Conference, Heidelberg, May 2024



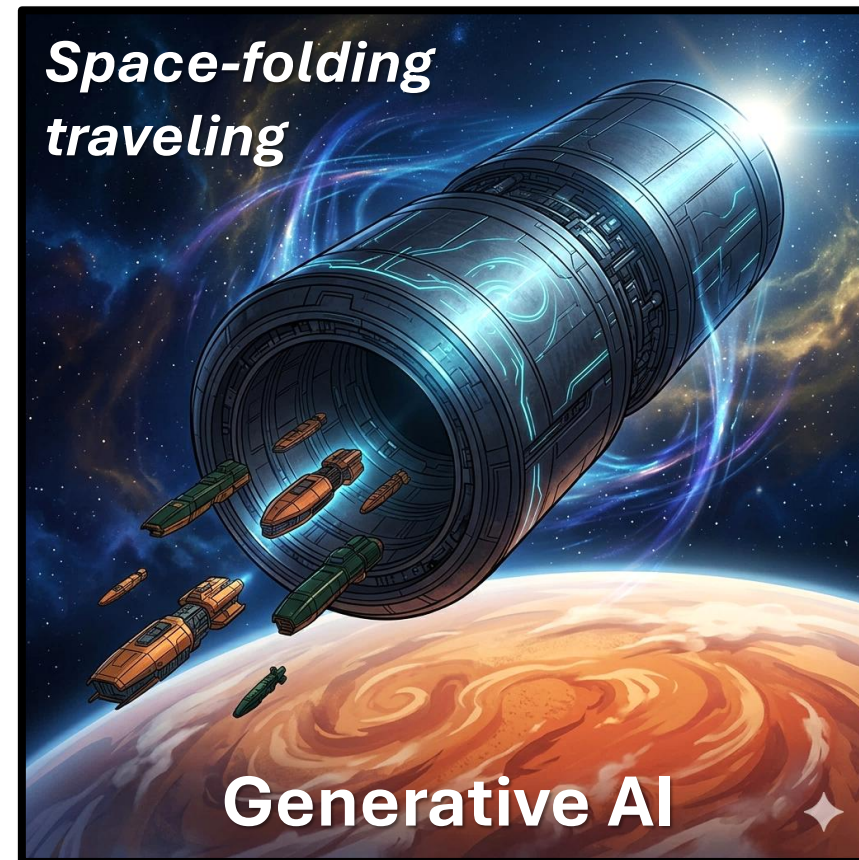
Chemical space traveling technology: StarWars *versus* Dune



Predictions for HTVS

$$X \rightarrow y$$

Find the best one we (can) have

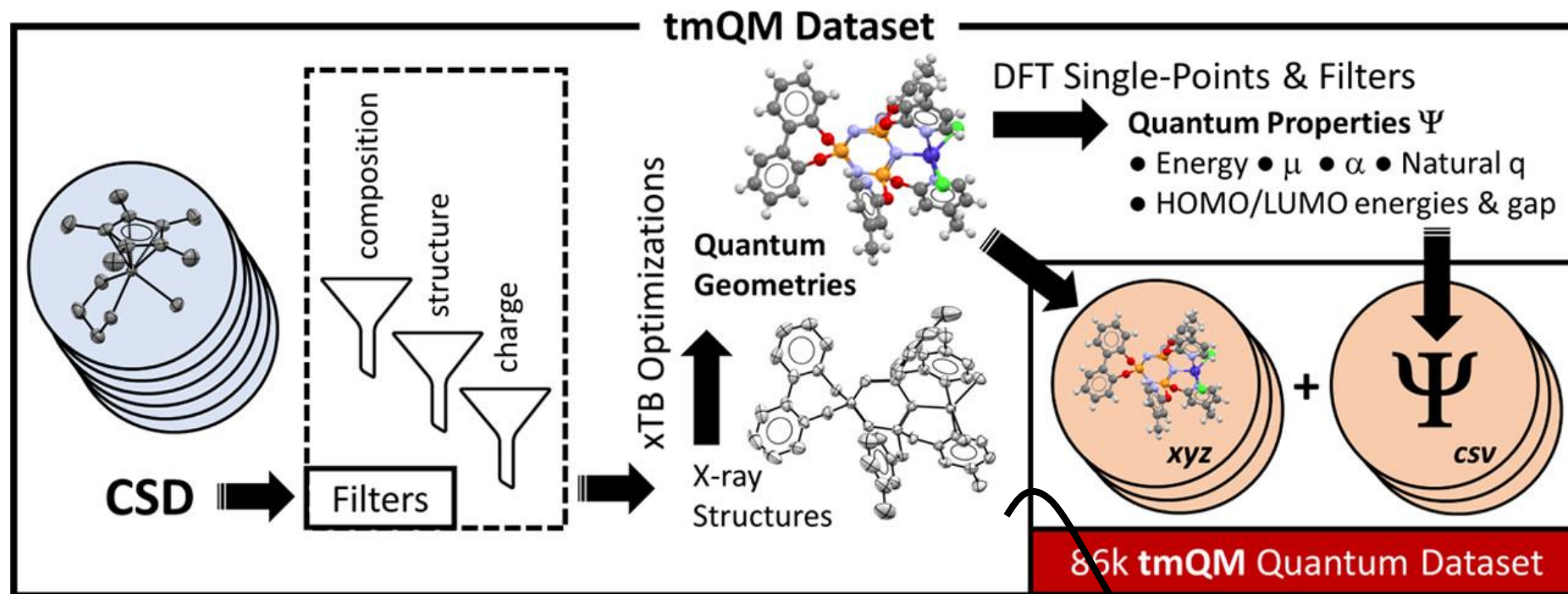


Inverse design

$$y \rightarrow X$$

Create the one we need

The tmQM dataset | tmQM = transition metal Quantum Mechanics



Bastian
Skjelstad

TMC = Transition Metal Complex

108k TMCs (2024 CSD)

Large and diverse: The 30 transition metals bound to >35K different ligands

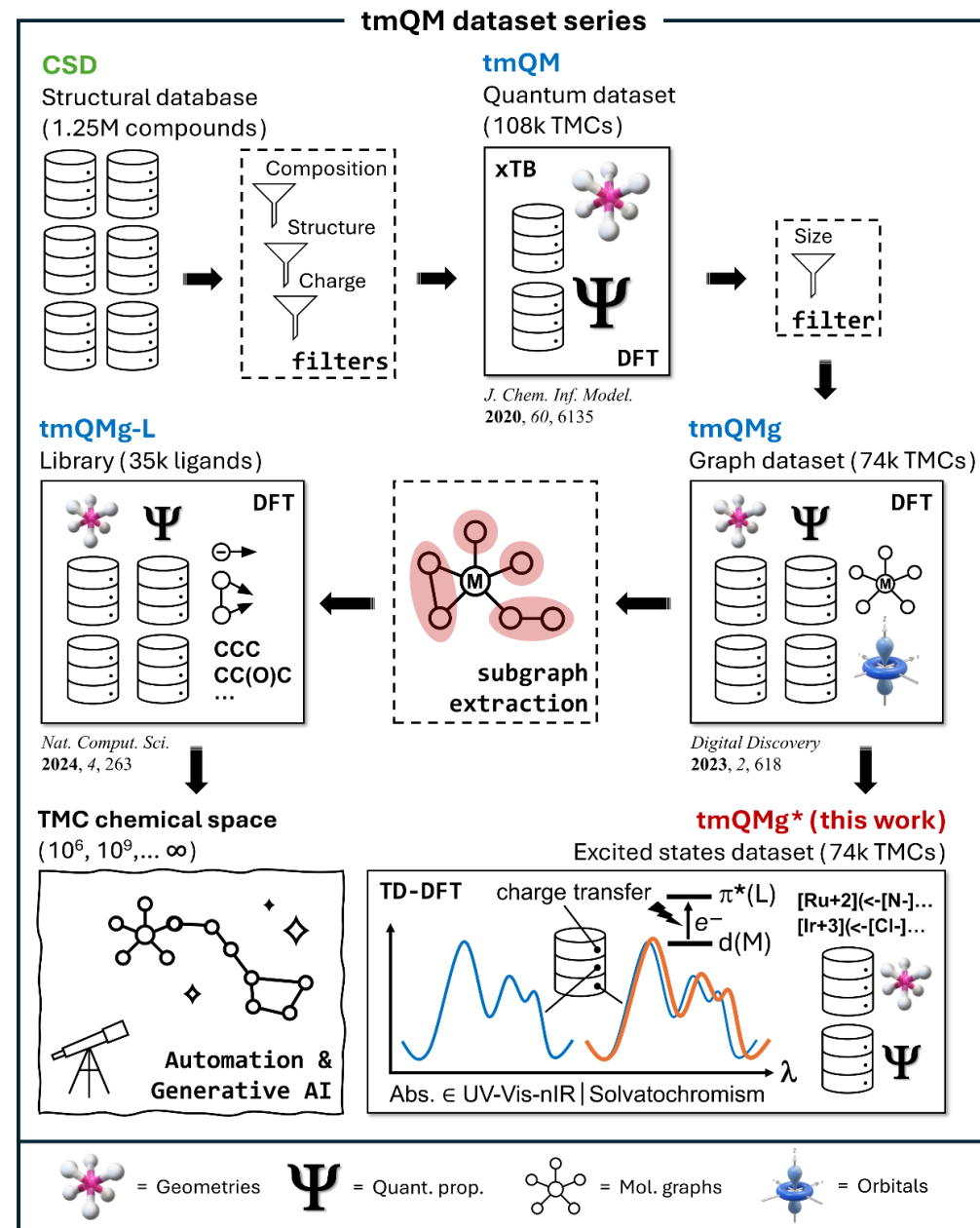
Machine learning: Valuable for training and evaluating ML models for metal complexes

Openly available: From the web pages of our group (uiocompcat.info), [GitHub](#) & [Zenodo](#)

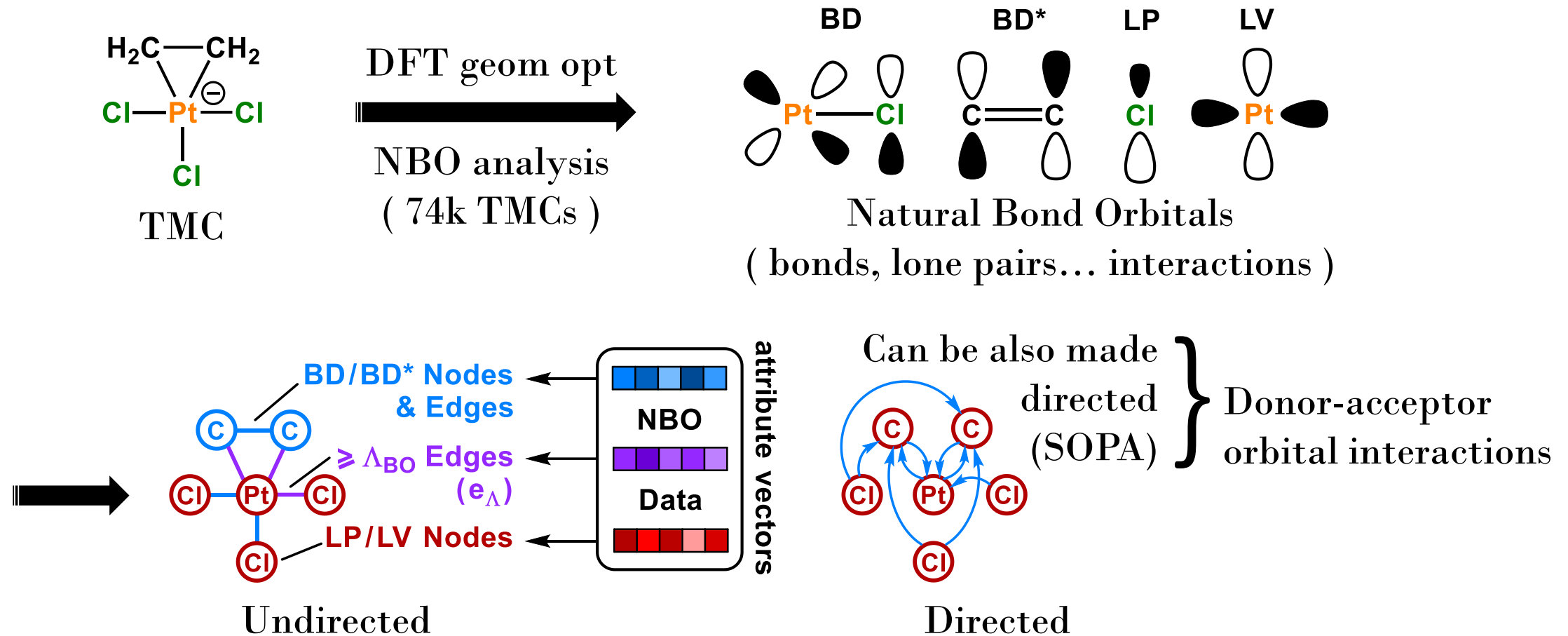
We turned **tmQM** into a dataset series:
tmQMg, **tmQMg-L**, **tmQMg***, **tmQM-RDF**, **tmQM-Δ**
 and then saw others contributing:



2020	tmQM	<i>Skjelstad, Balcells</i>
2023	ReaLigands	<i>Ess</i>
2023	tmQM-ωB97MV	<i>Ulissi, Blau</i>
2025	tmBIO, tmCAT...	<i>Kulik</i>
2025	tmQM+	<i>Alexandrova</i>
2025	OMol25	<i>Meta</i>



The graph dataset, tmQMg, was key in developing the series



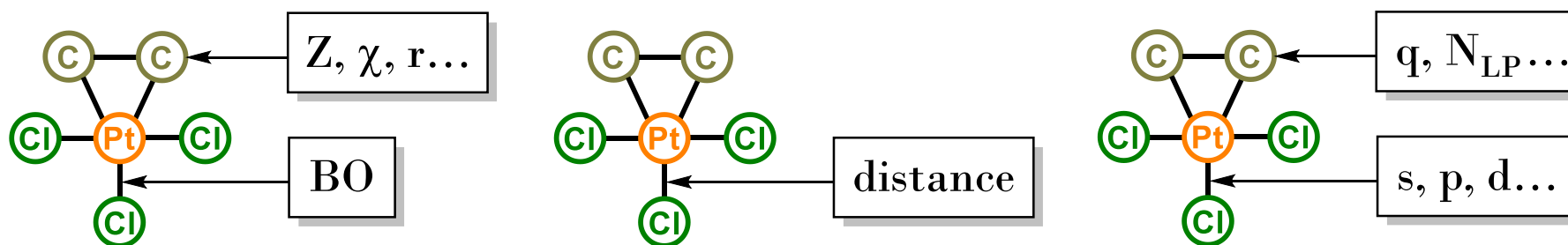
Natural quantum graphs: NBO topology & attribution $\Rightarrow$ GNN predictions

Kneiding, Balcells, and co-workers, *Digital Discovery*, **2023**, 3, 618

Highlighted in *Nature Computational Science*

Leveraging tmQMg in the prediction of quantum properties

Electronic structure information more valuable than the geometric



periodic properties $\Rightarrow$ $\oplus$ *geometric (d)* $\Rightarrow$ $\oplus$ *electronic (NBO)*

Decrease in mean absolute errors (GNN):

Periodic		Geometric		Electronic	
2.0	$\xrightarrow{-0.3}$	1.7	$\xrightarrow{-0.8}$	0.9	Dipole moment
9.0	$\xrightarrow{-0.7}$	8.3	$\xrightarrow{-2.3}$	6.0	HOMO-LUMO gap
6.4	$\xrightarrow{-0.5}$	5.9	$\xrightarrow{-1.0}$	4.9	Polarizability

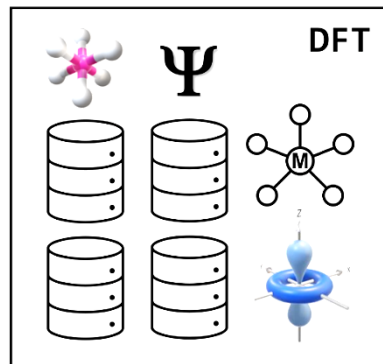
Kneiding, Balcells, and co-workers, *Digital Discovery*, **2023**, 3, 618

Highlighted in *Nature Computational Science*

Last addition to the series: tmQMg*

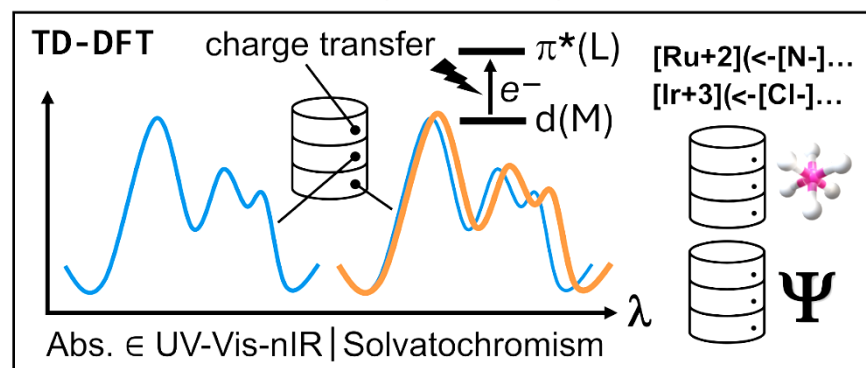
tmQMg

Graph dataset (74k TMCs)



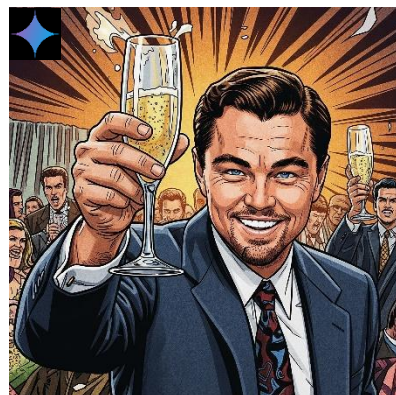
tmQMg*

Excited states dataset (74k TMCs)

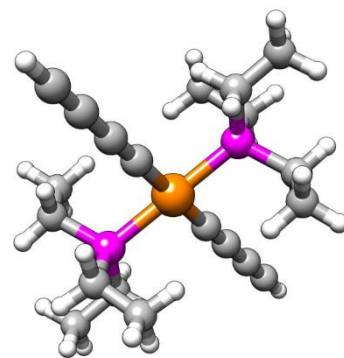
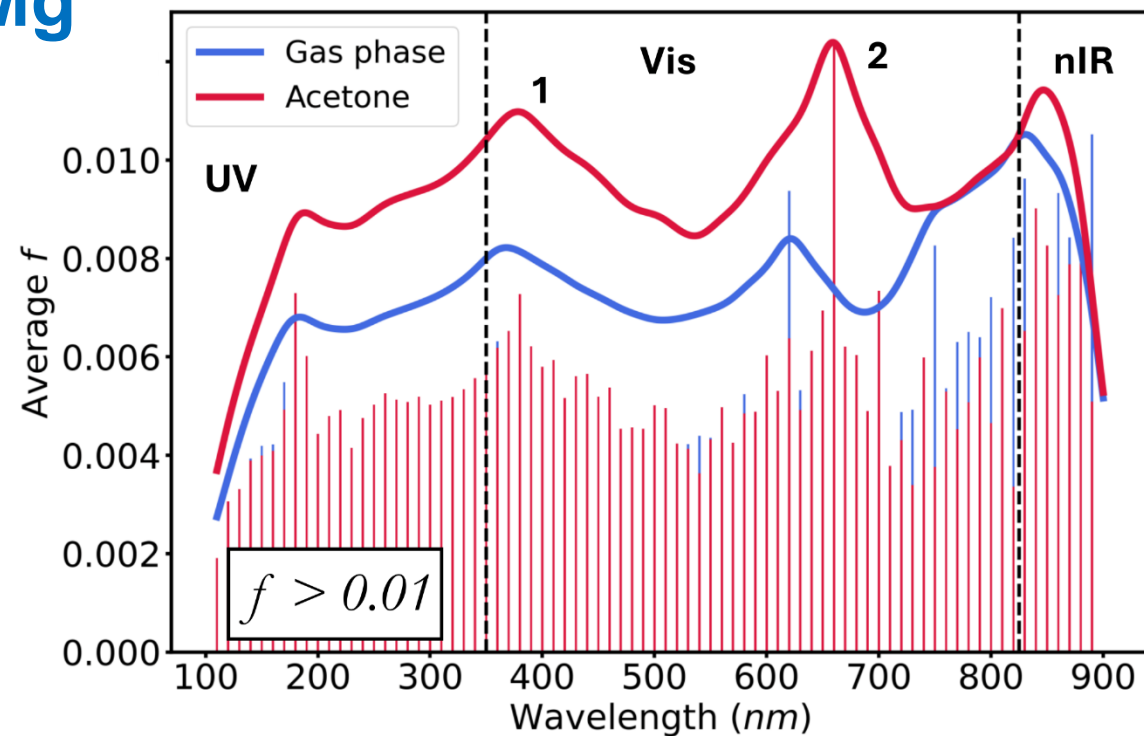


Wavelengths & intensities (ωB97xd/def2SVP)

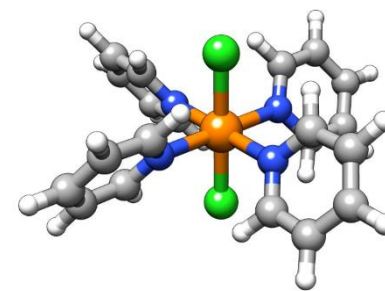
Next? Spin & H⁺ states
(electrochemistry)



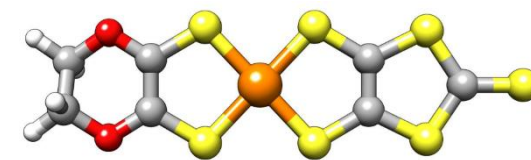
This is not for free
tmQM ≡ 15M CPU h



FEGJOR, Pt(II), UV



TIRGOQ, Ti(II), Vis 2

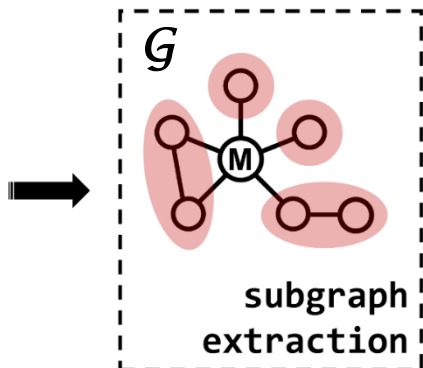
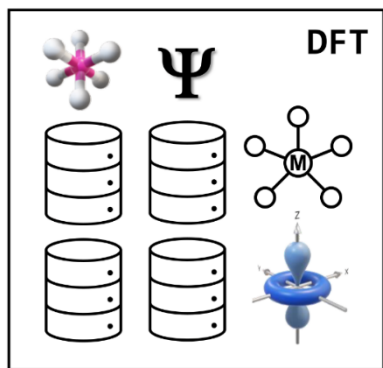


ICIHUX, Ni(IV), nIR

Building a large ligand library: tmQMg-L

tmQMg

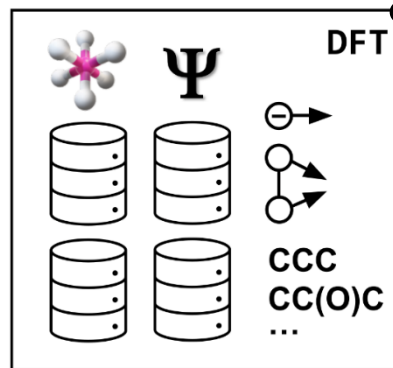
Graph dataset (74k TMCs)



NBO $\rightarrow$ L charge
topology $\rightarrow$ L coordination

tmQMg-L

Library (35k ligands)

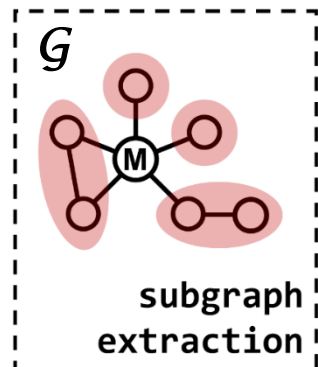
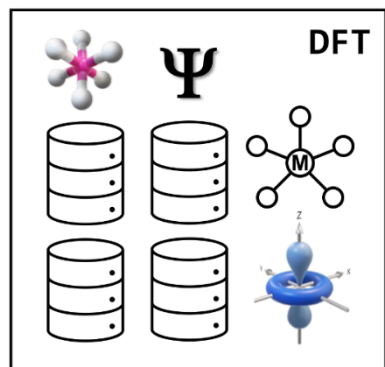


Extensive Chemoinformatics from
RDKit and Jorner's MORFEUS
(SMILES, logP, steric parameters...)

Building a large ligand library: tmQMg-L

tmQMg

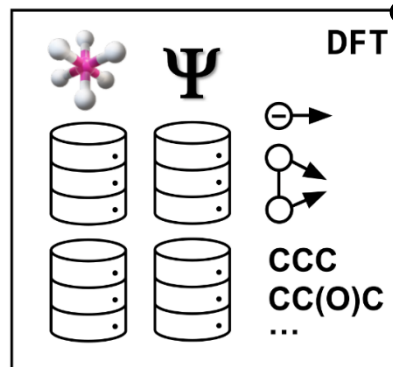
Graph dataset (74k TMCs)



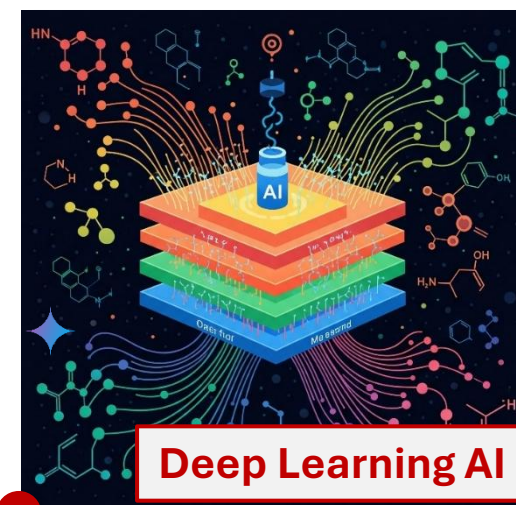
NBO $\rightarrow$ L charge
topology $\rightarrow$ L coordination

tmQMg-L

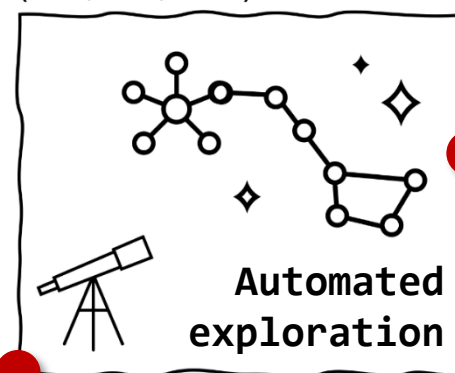
Library (35k ligands)



Extensive Chemoinformatics from
RDKit and Jorner's MORFEUS
(SMILES, logP, steric parameters...)

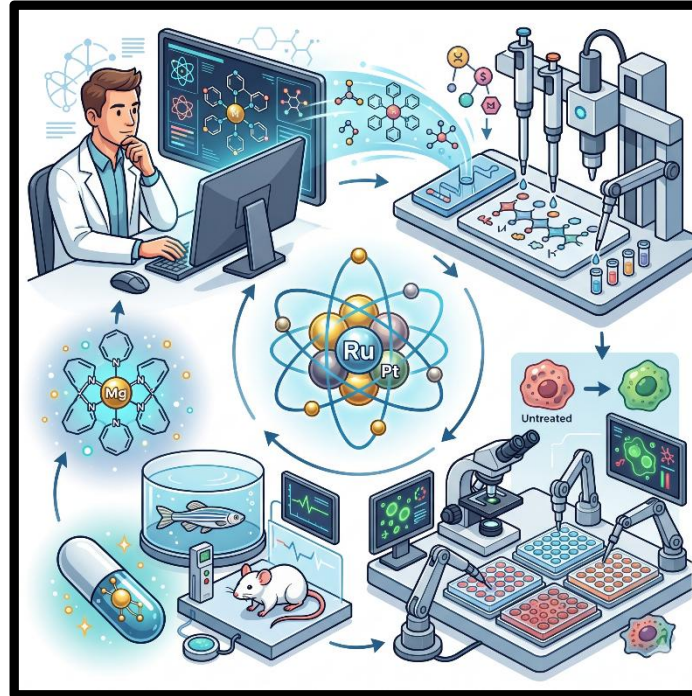


TMC chemical space
(10^6 , 10^9 , ... ∞)



Generative AI

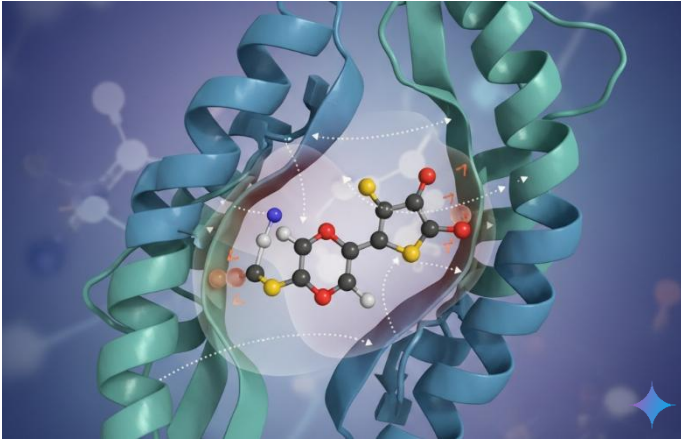
Why are we interested in the chemistry of the transition metals?



METALLODRUG DISCOVERY

Multiobjective optimization problems: Genetic algorithms

Strongly interacting, large α



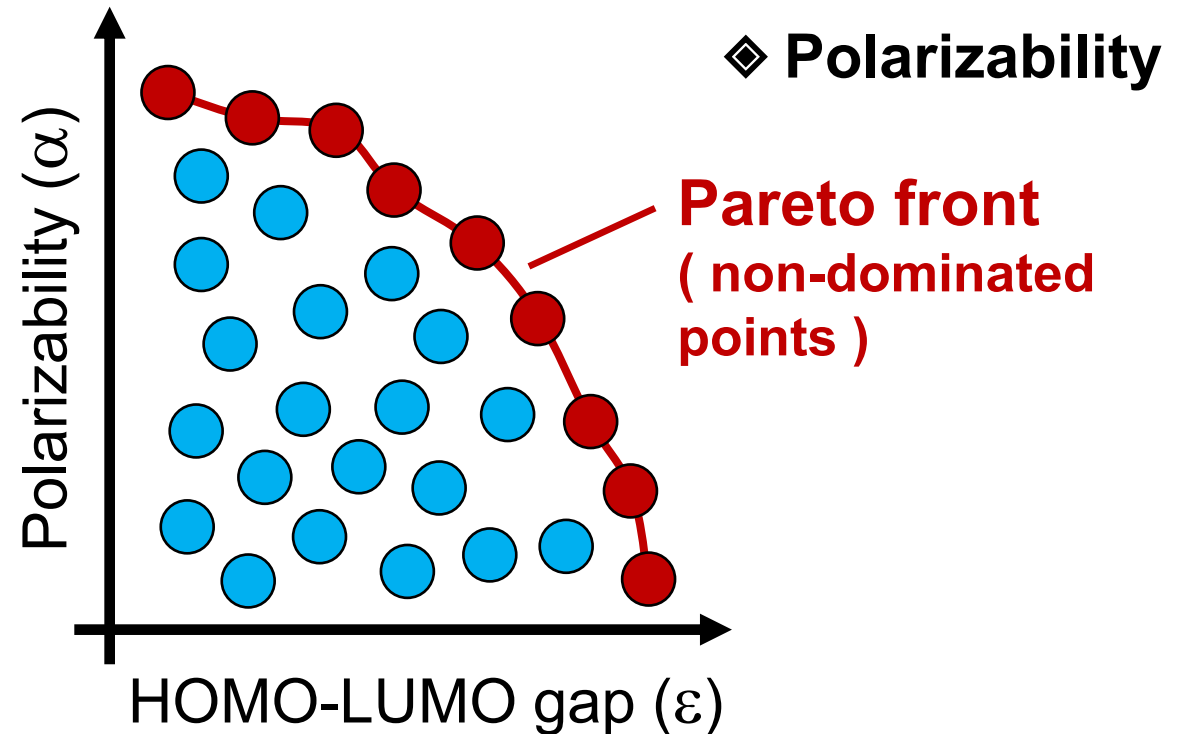
Highly stable, wide ϵ



Benchmark toy task: Joint optimization of

◆ HOMO-LUMO gap

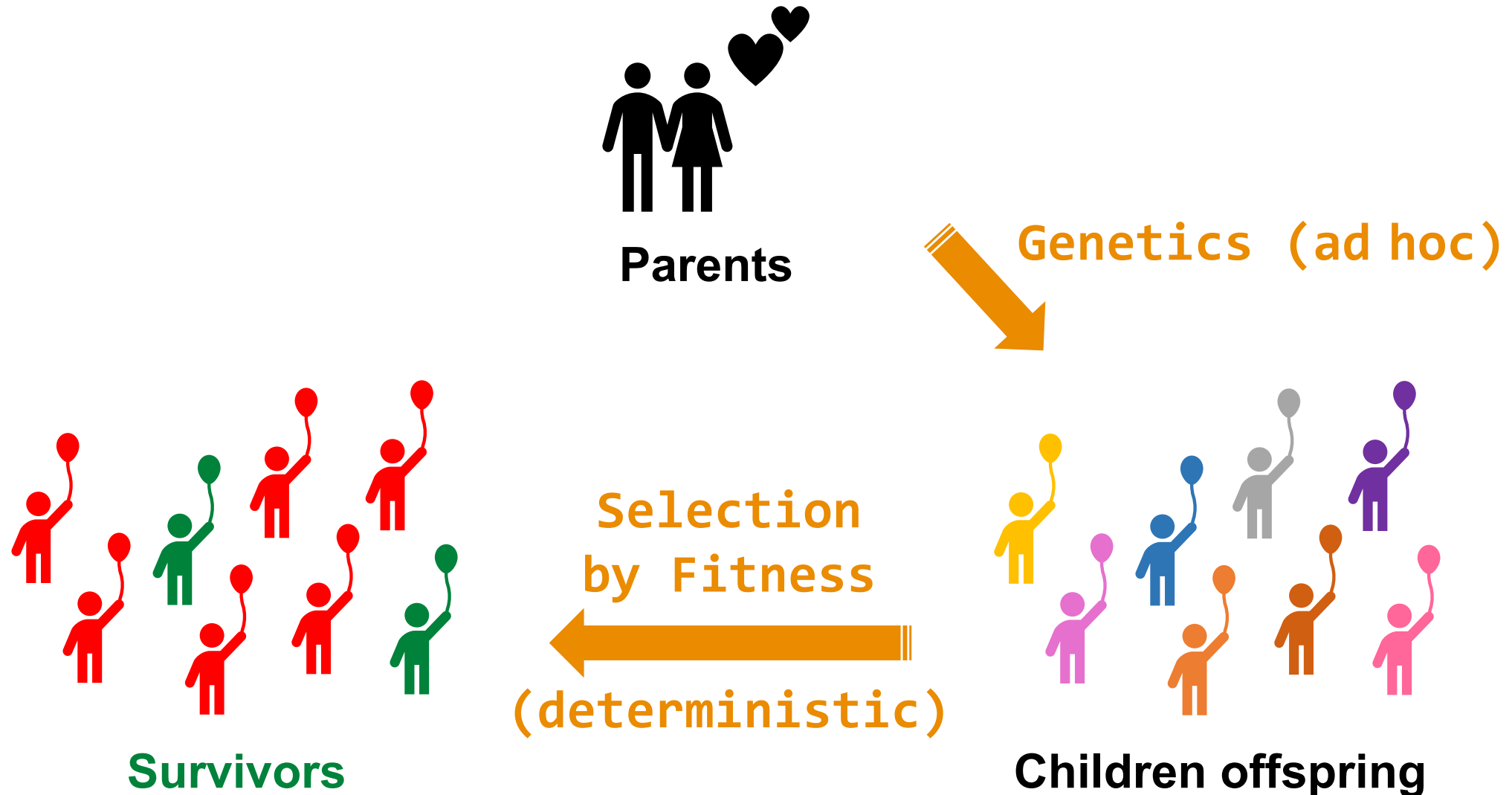
◆ Polarizability



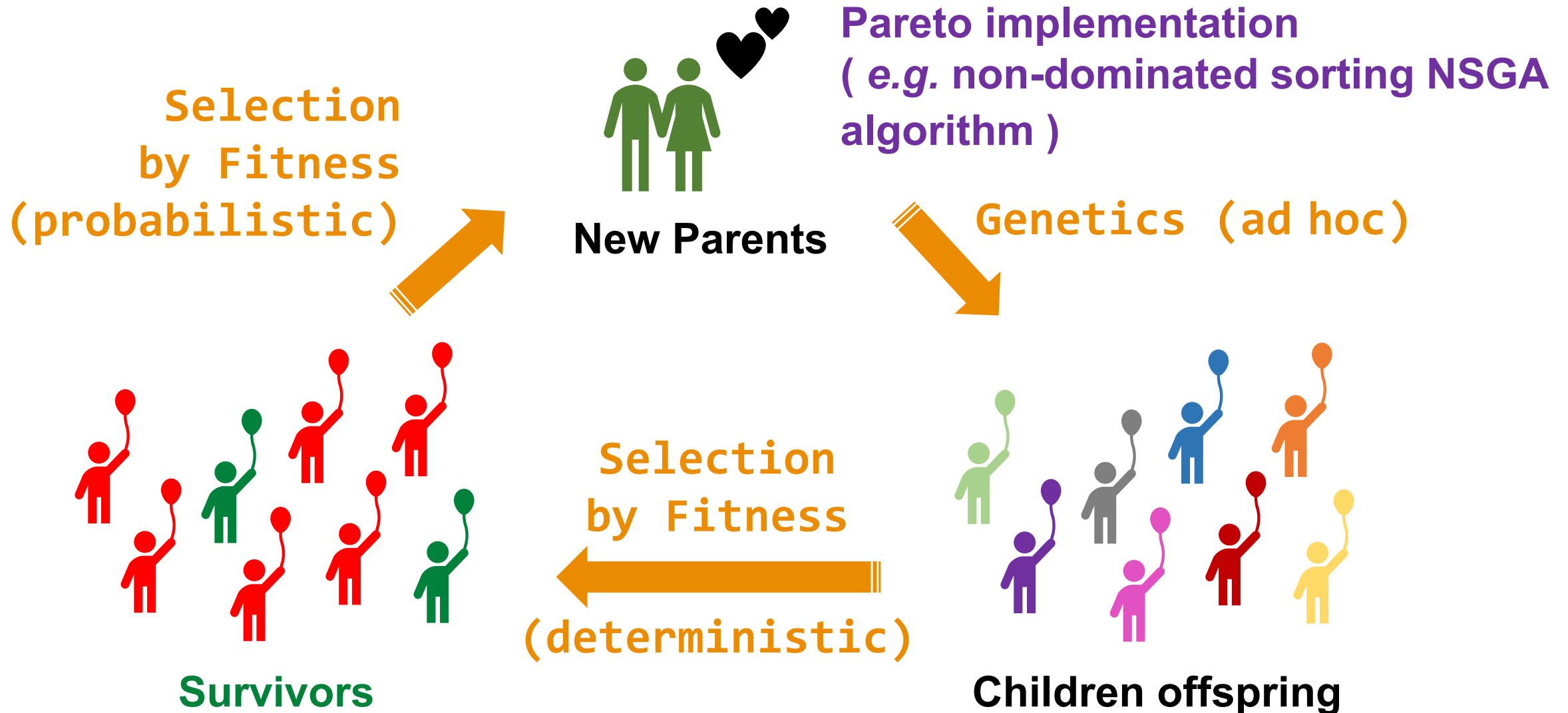
NaviCatGA (Laplaza, Gallarati, Corminboeuf)

DENOPTIM (Foscatto, Venkatraman, Jensen)

Multiobjective optimization problems: Genetic algorithms

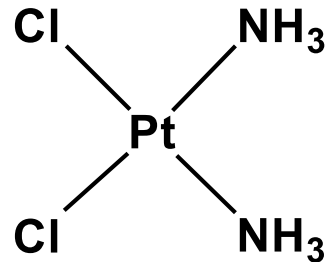


Multiobjective optimization problems: Genetic algorithms

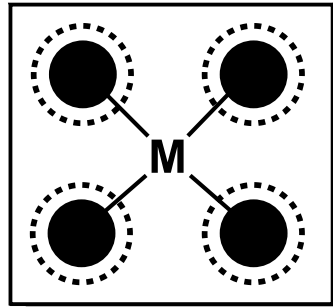


Genetic operations for metal complexes

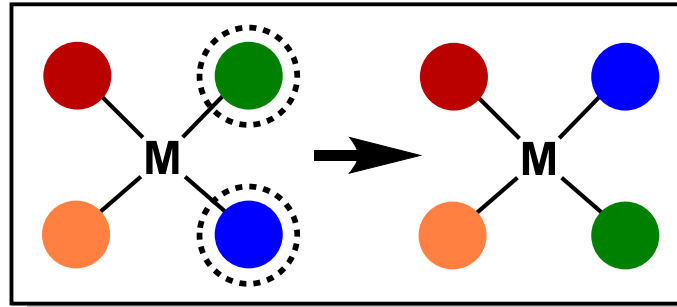
cisplatin



Chromosome:



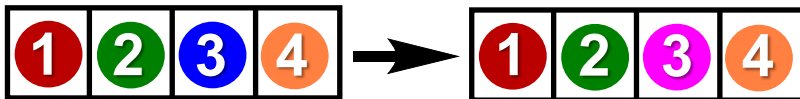
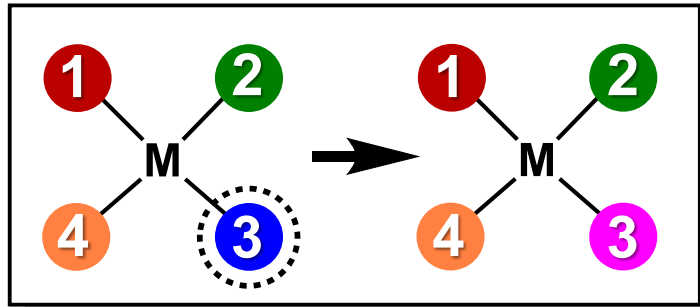
Two-ligand mutation (swap; cis/trans):



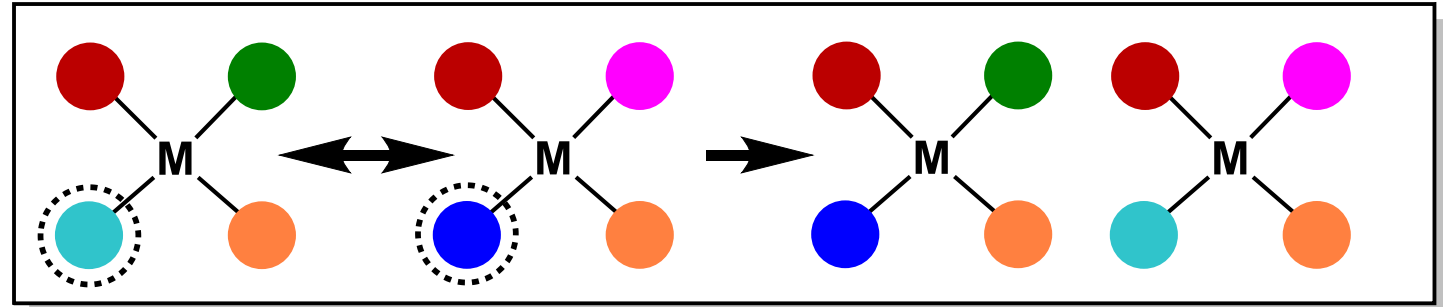
Genes (metal ligands):



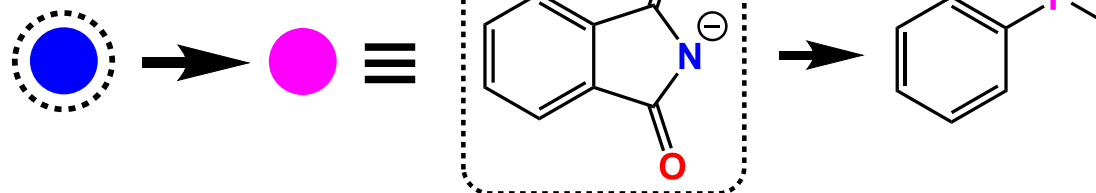
Single-ligand mutation:



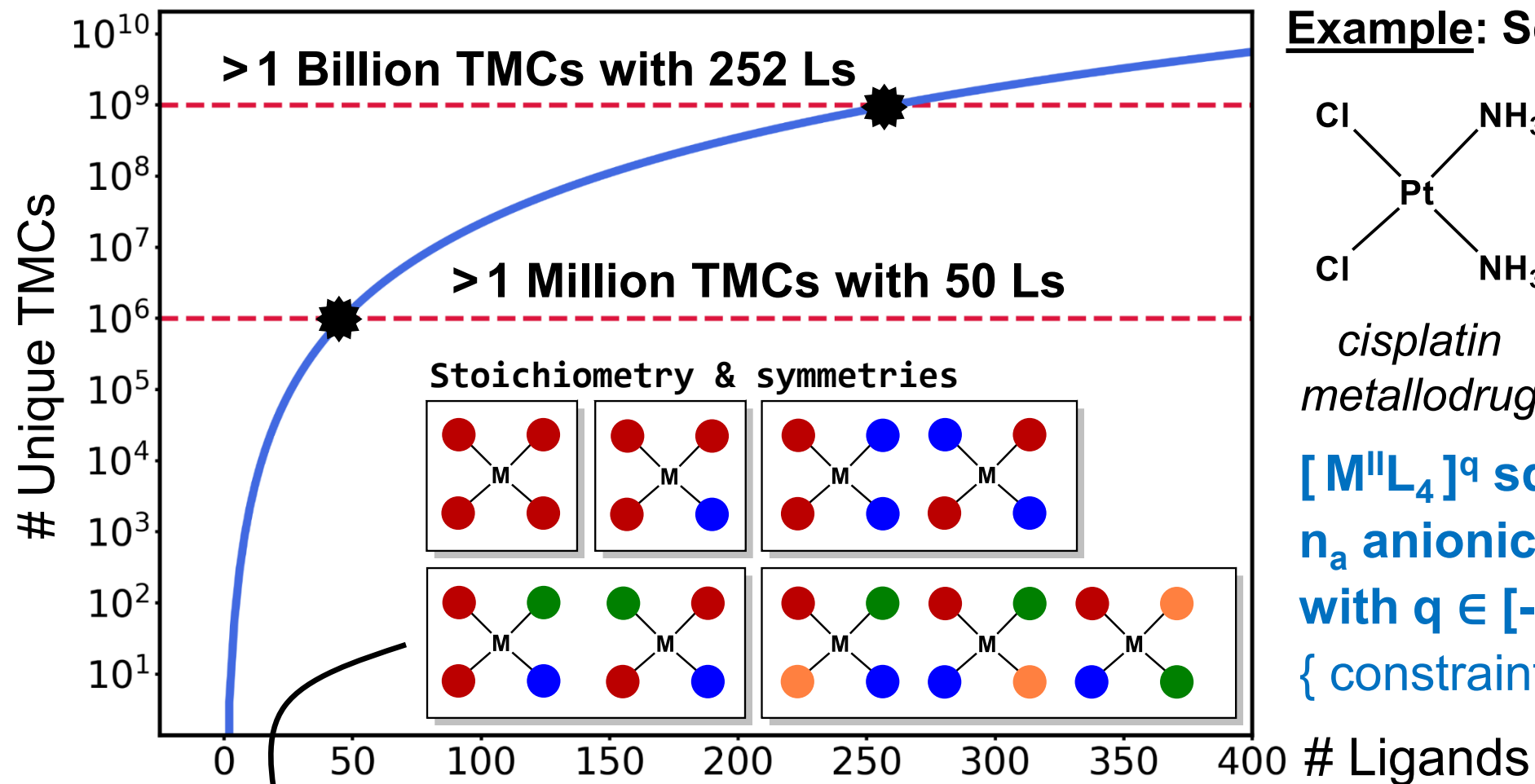
Crossover (ligand-exchange):



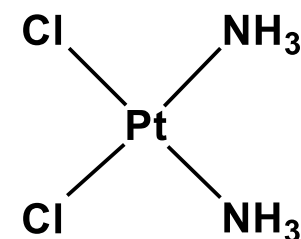
Full-ligand operations; e.g.:
(L pools from tmQMg-L)



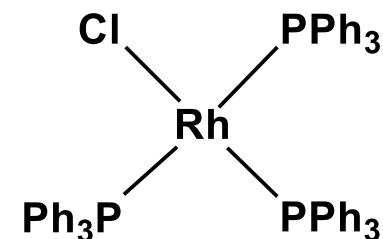
But how large are these spaces? They quickly become vast!



Example: Square planar TMCs



cisplatin
metallodrug

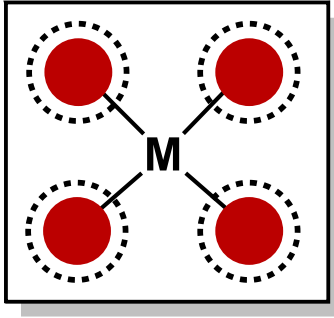


Wilkinson's
catalyst

$[M^q L_4]$ square planar space
 n_a anionic, n_n neutral κ^1 -Ls,
 with $q \in [-1, 0, +1]$
 { constraints }

$$N_{TMC} = n_a^3 n_n + \frac{3}{2} n_a^2 n_n^2 + n_a n_n^3 + \frac{1}{2} n_a n_n$$

Finding a needle in the solar system



All κ^1 ligands in tmQMg-L (CSD)

$n_n = 6,503$ neutral ligands

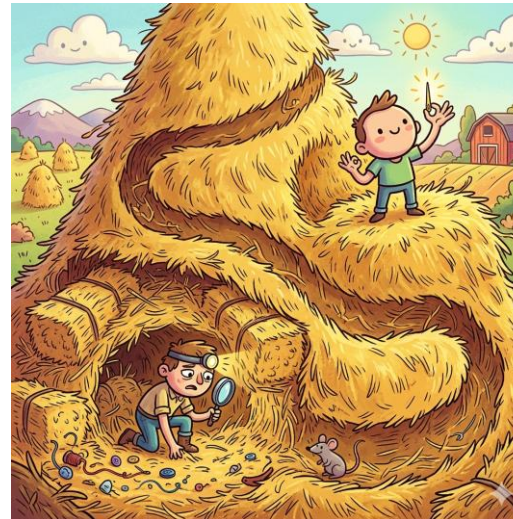
$n_a = 2,864$ anionic

Pd(II) complexes with four κ^1 ligands, $\{-1, 0, +1\}$ charge

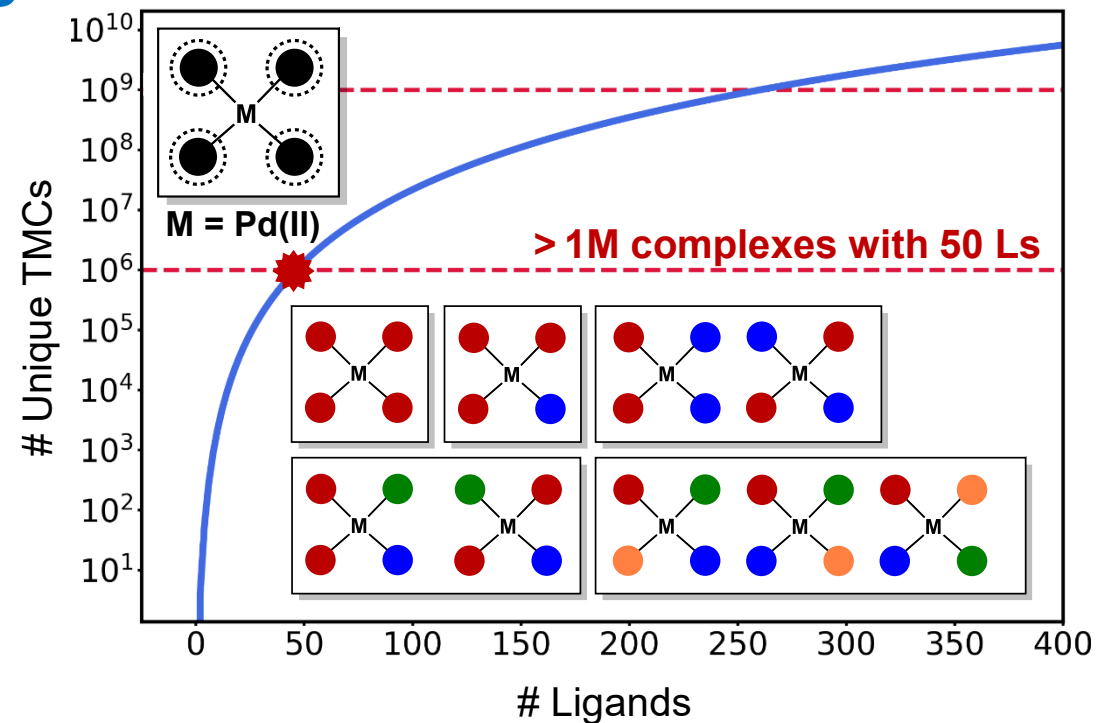
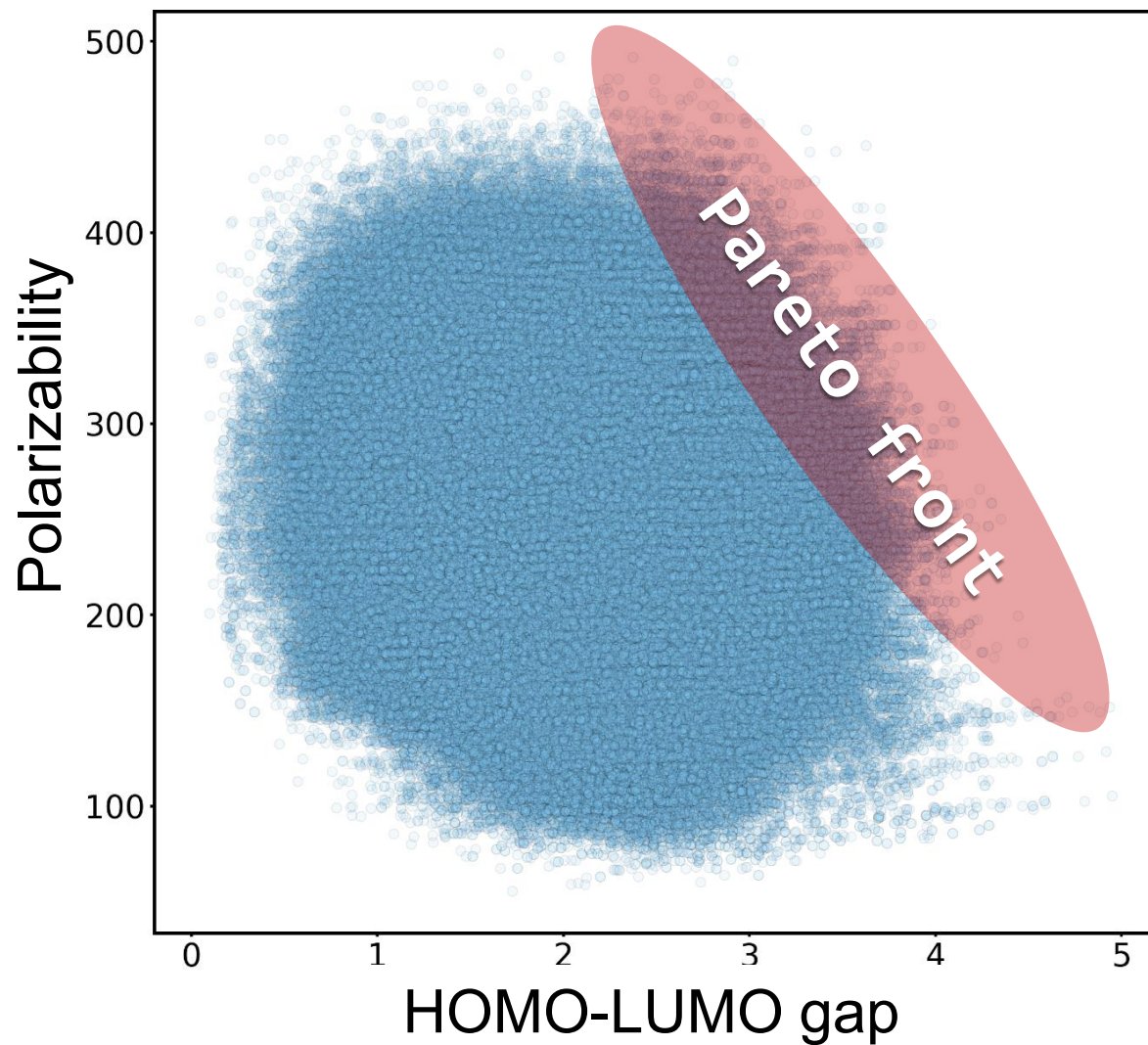
$$N_{TMC} = n_a^3 n_n + \frac{3}{2} n_a^2 n_n^2 + n_a n_n^3 + \frac{1}{2} n_a n_n$$

$$N_{TMC} = 1.46 \times 10^{15}$$

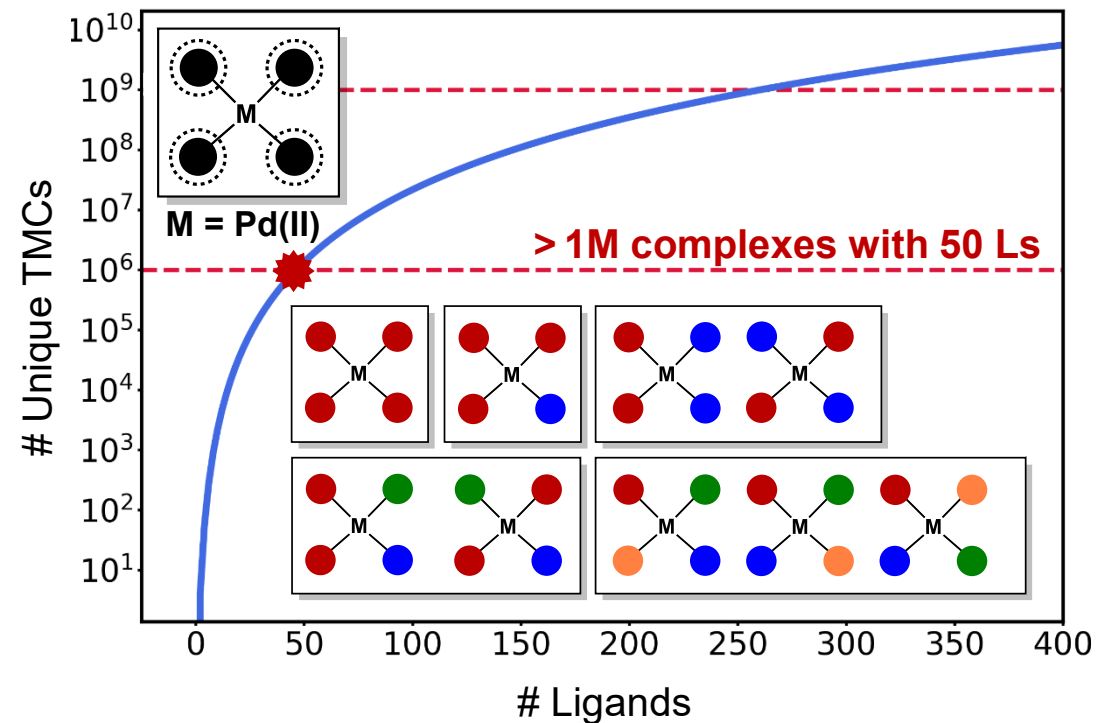
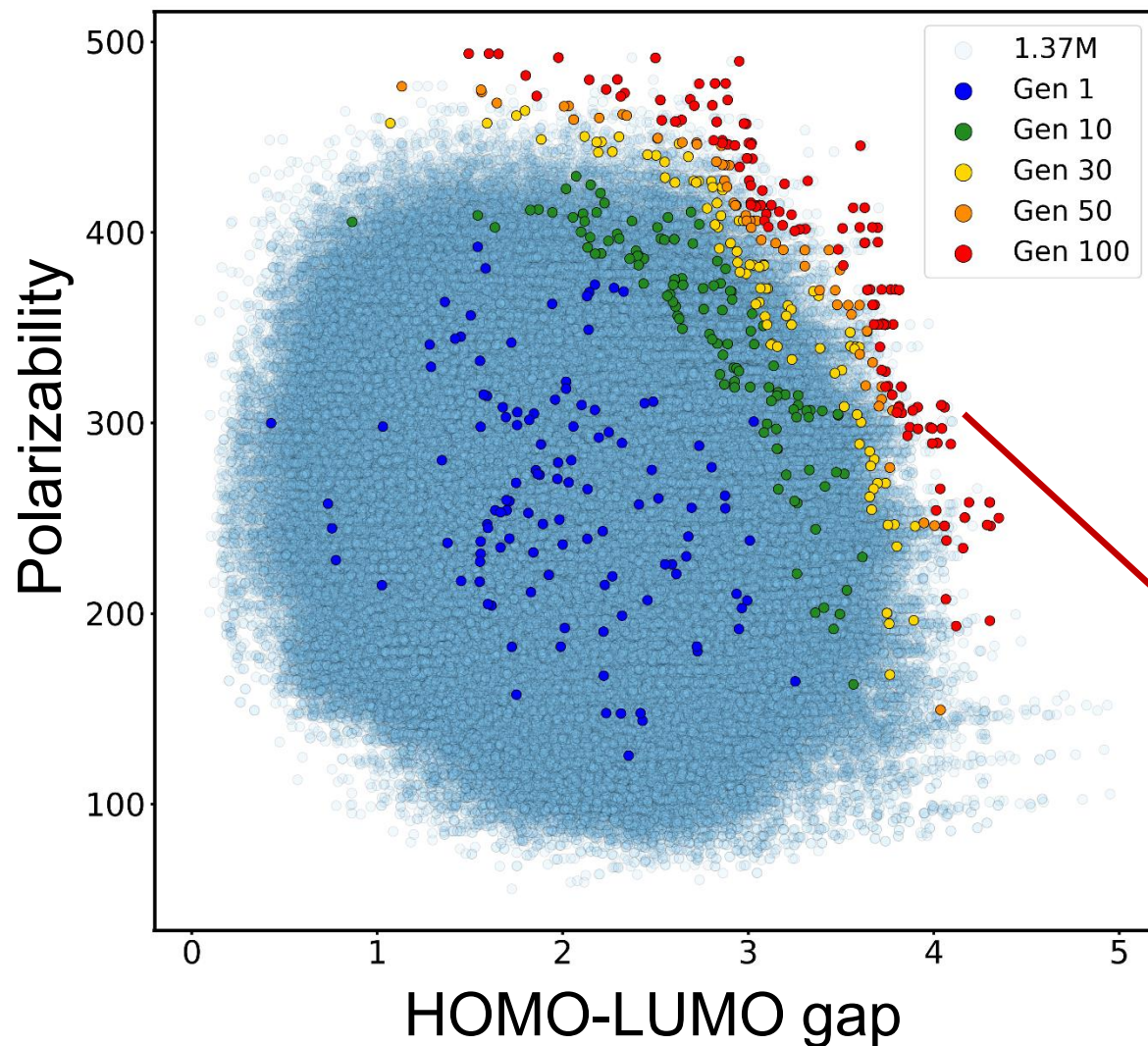
Sun to Pluto distance =
 0.59×10^{15} cm



Benchmark space: 1.37M PdL₄ TMCs



Pareto optimization in 1.37 M space

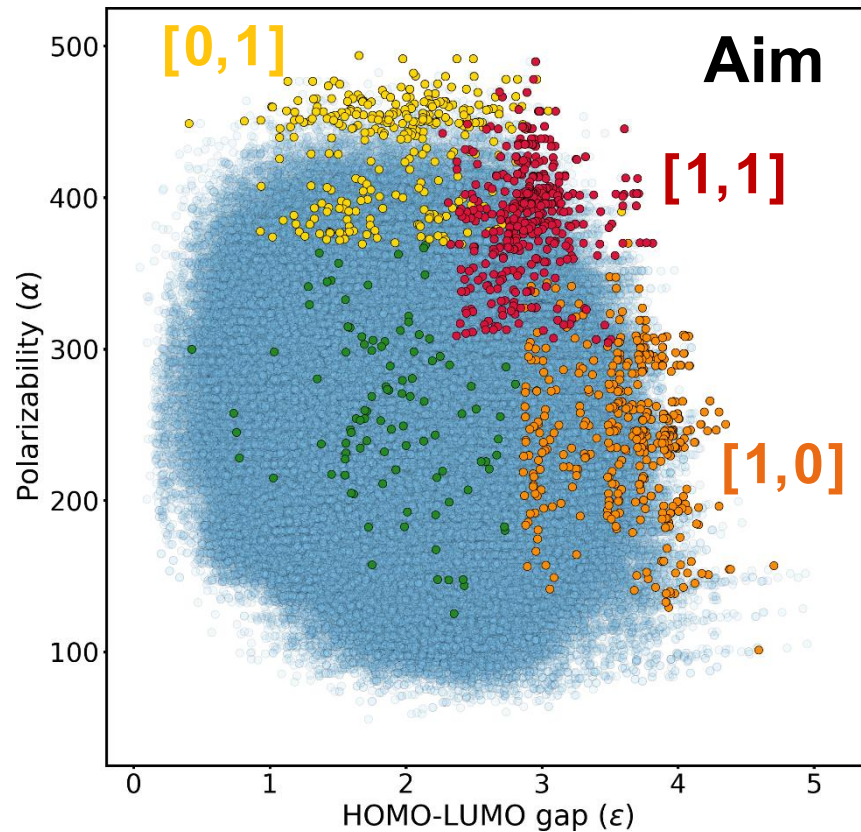


**Found 130 Pareto front solutions,
18/30 non-dominated points**

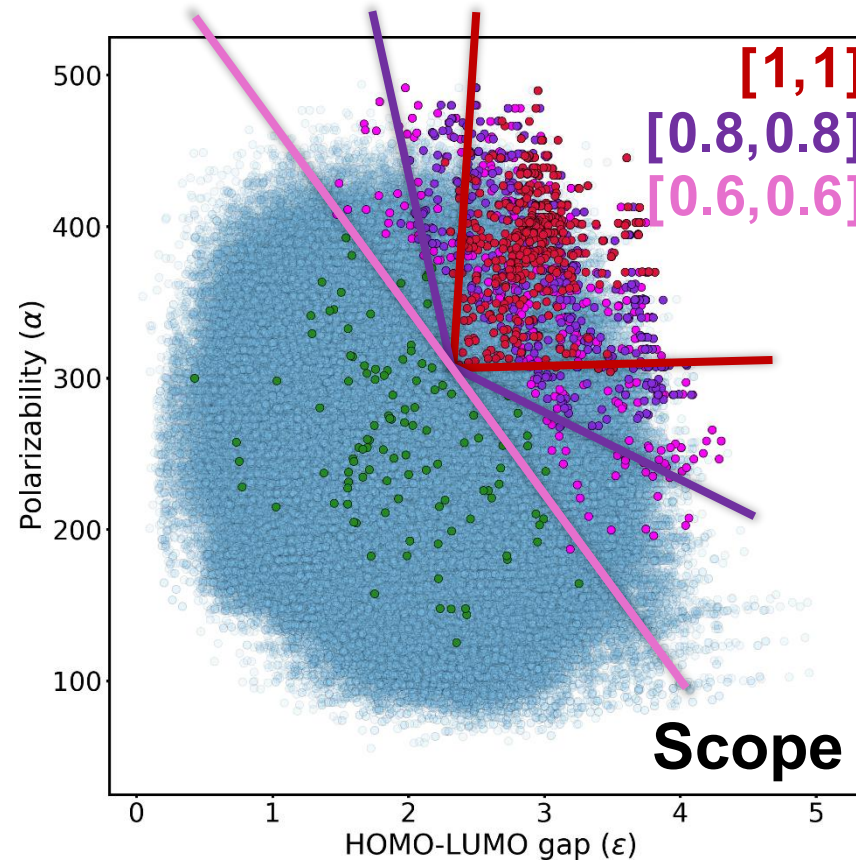
Highly efficient optimization:

Only 1% of the global space was explored

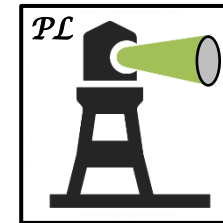
PL-MOGA algorithm: Directional multiobjective optimization



- Maximum polarizability
- Maximum gap
- Maximum both



- Narrow scope
- Medium scope
- Wide scope
- { both properties maximized }



PL-MOGA:
NSGA with a
masking function:

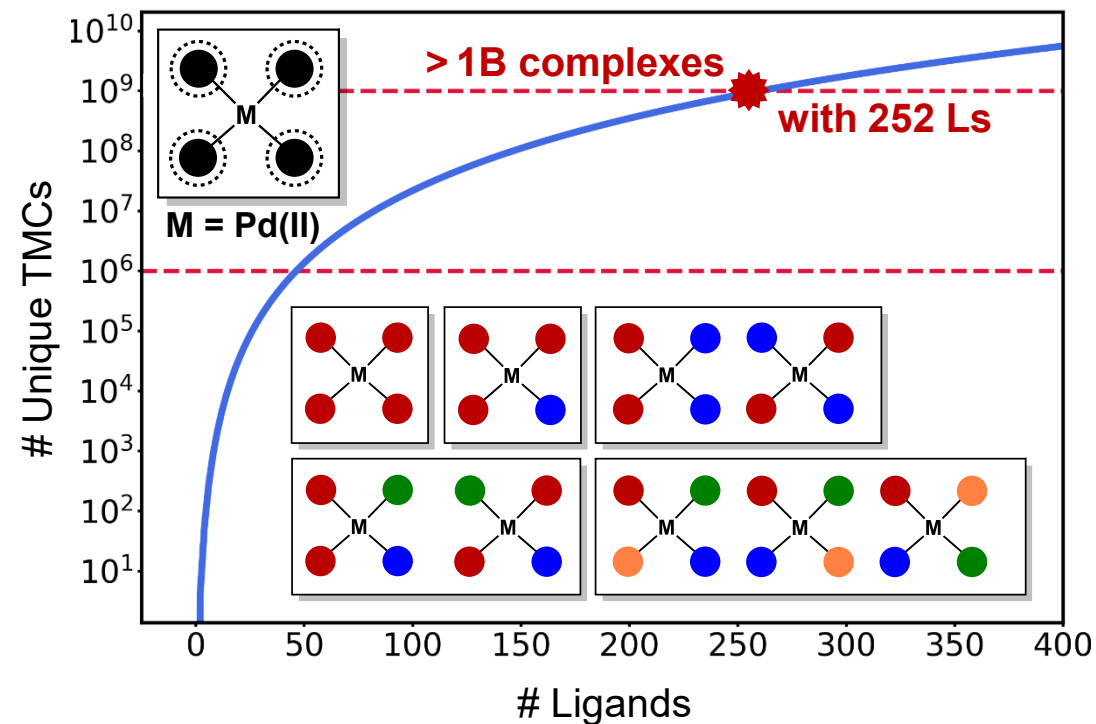
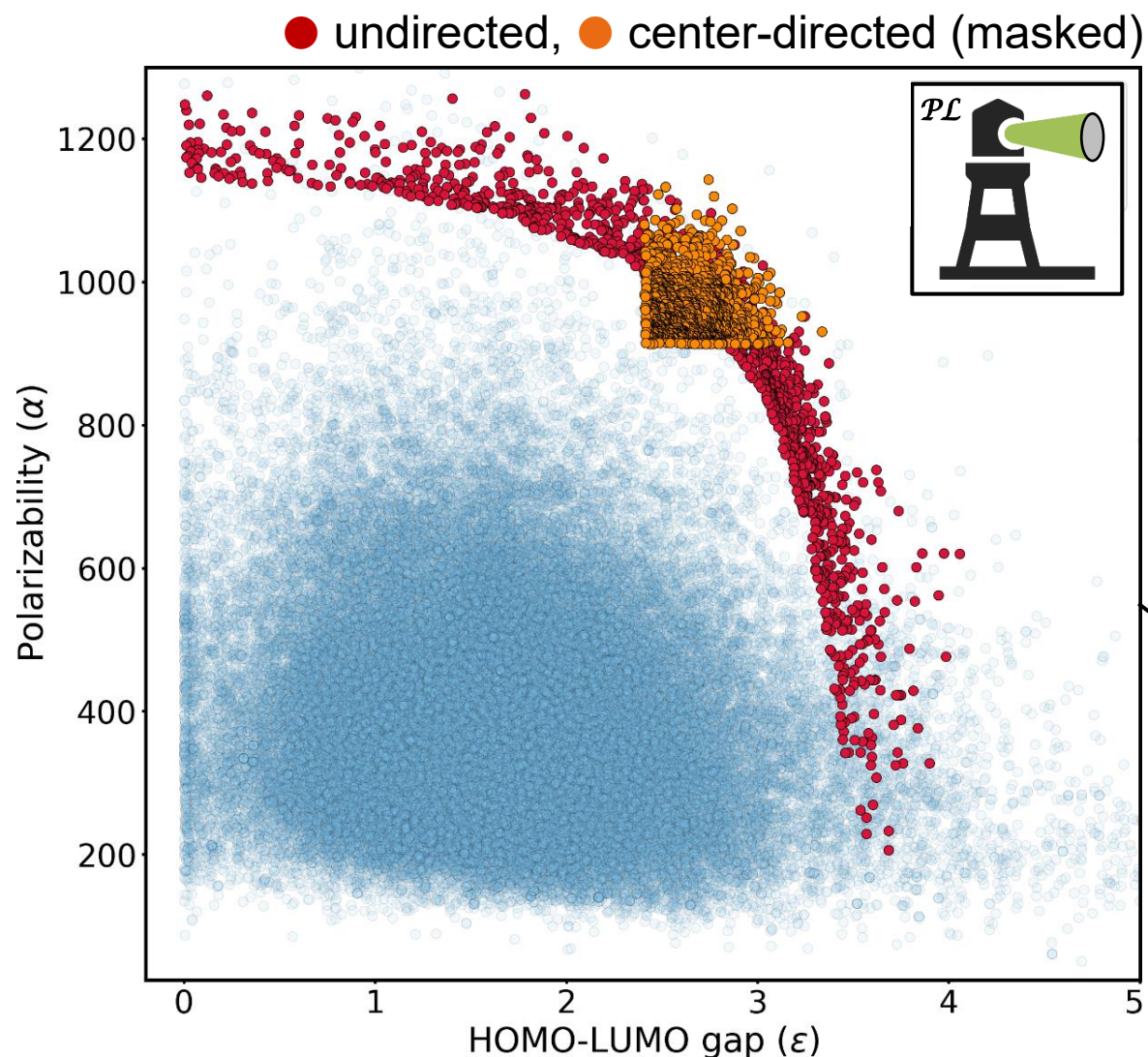
```

for t in T do
   $m_t \leftarrow t$  median pop  $x$ 
  if  $x_t < s_t \cdot m_t$  then
    return  $x_t \leftarrow 0$ 
  end if
end for
[  $0 \leq s_t \leq 1$  ]
    
```

Better than weighted-sum:

- 1) No prior knowledge
- 2) Finer control

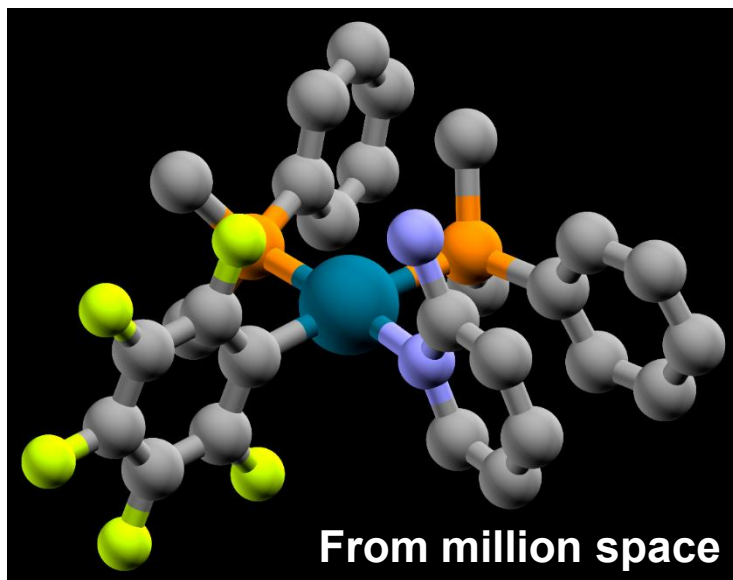
Billion-system space



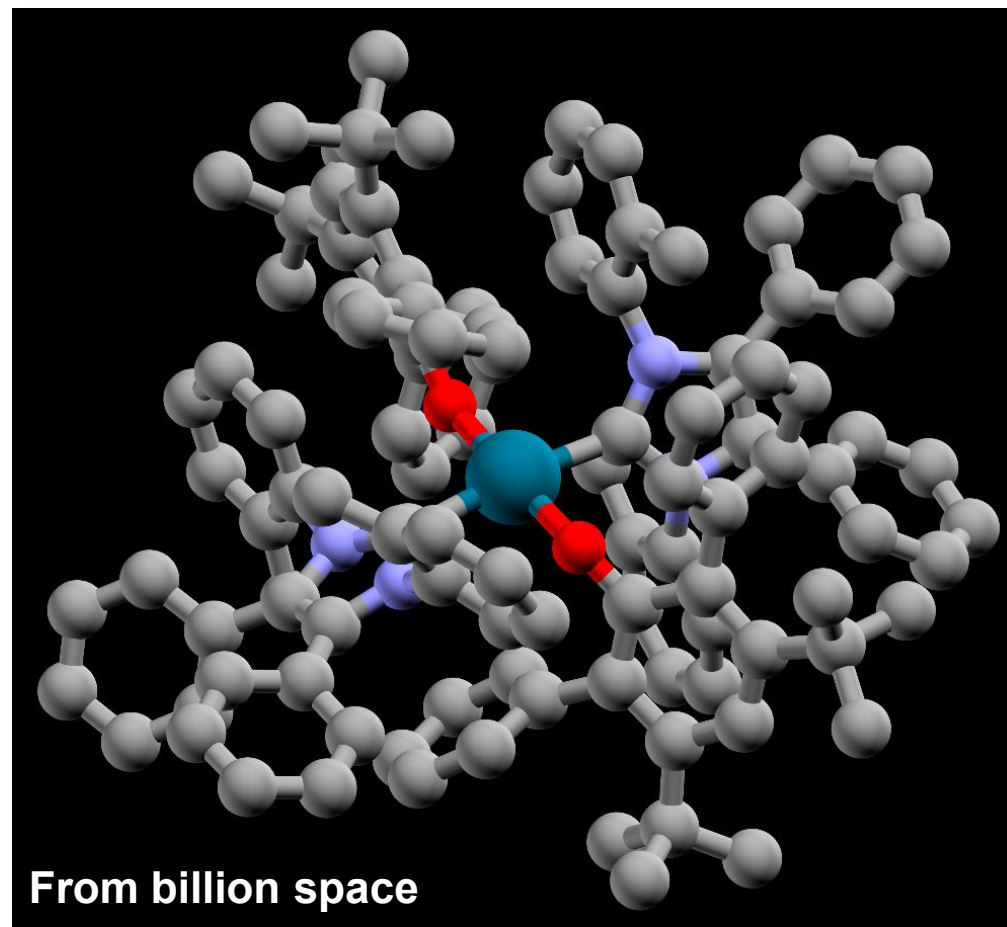
**1 300 Pareto front TMCs evolved
using 252 random ligands**

The reference background (●) is tmQM; *i.e.*,
a large (86k) sample of the CSD database,
combining the 30 TMs with 30k ligands

Pareto complexes evolved by the PL-MOGA genetic algorithm



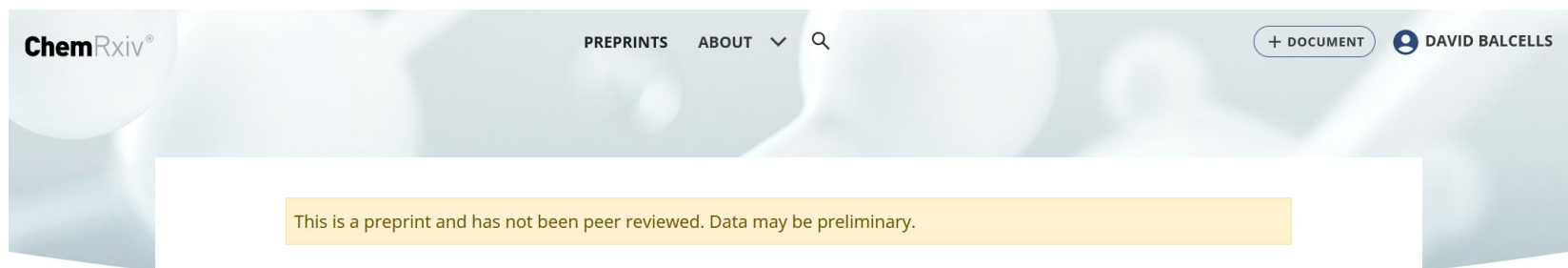
Highly diverse TMCs with Pareto properties, makeable ligands & 3D structures available



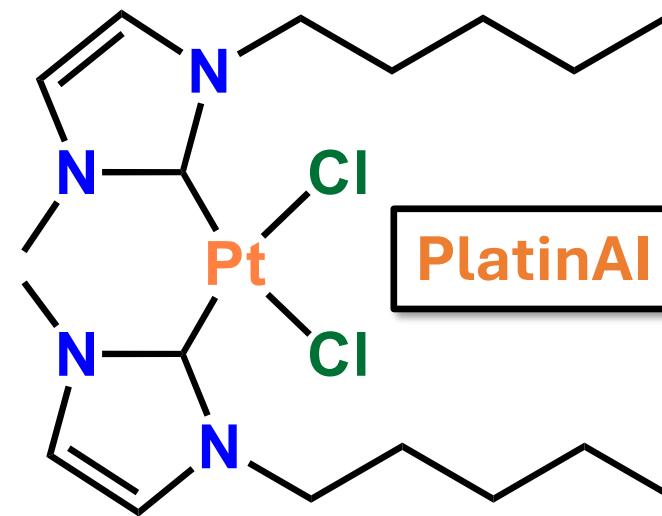
Structure assembly: molSimplify
Polarizability and gap: GFN2-xTB calculations
Geometry verification of elite TMCs: DFT(PBE/def2SVP)

(Further onto metallodrug discovery – Experimentally verified ML)

KANO: Experimentally tested on 19 platinum complexes active & inactive in anticancer therapy (83-93% accuracy), screens ~200k chemical space finding **PlatinAI**, a potent metallodrug that overcame cisplatin resistance both *in vitro* and *in vivo* via a different mechanism of action.



Manually curated dataset:
1k papers, 3.7k Pt TMCs,
17.7k IC₅₀ values



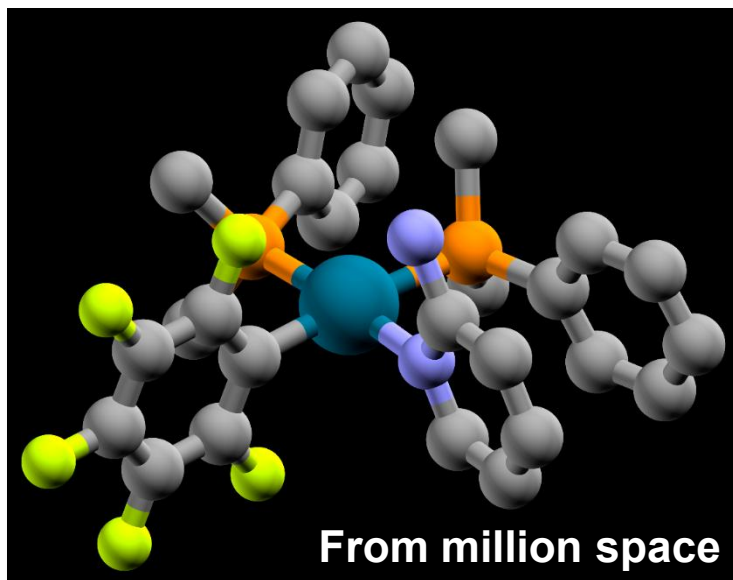
Prof. Maria Babak

Daniil Rusanov

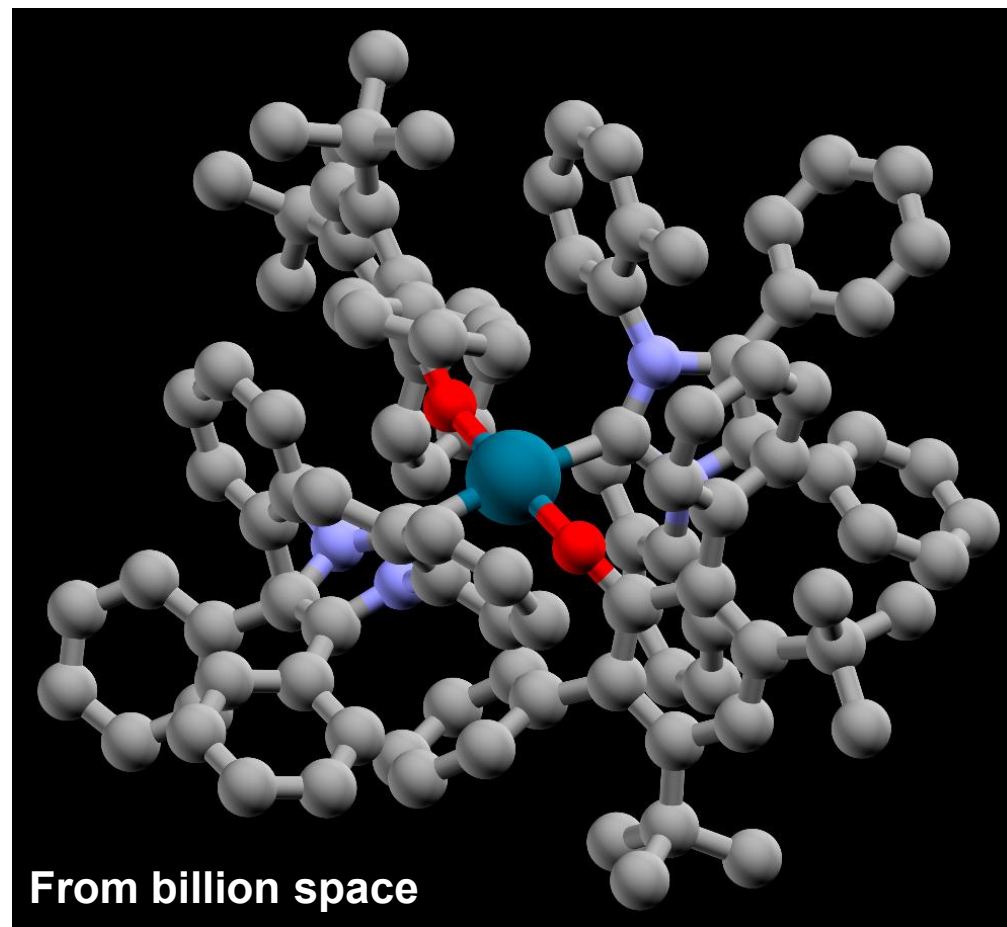
City University of Hong Kong



Pareto complexes evolved by the PL-MOGA genetic algorithm



Highly diverse TMCs with Pareto properties, makeable ligands & 3D structures available

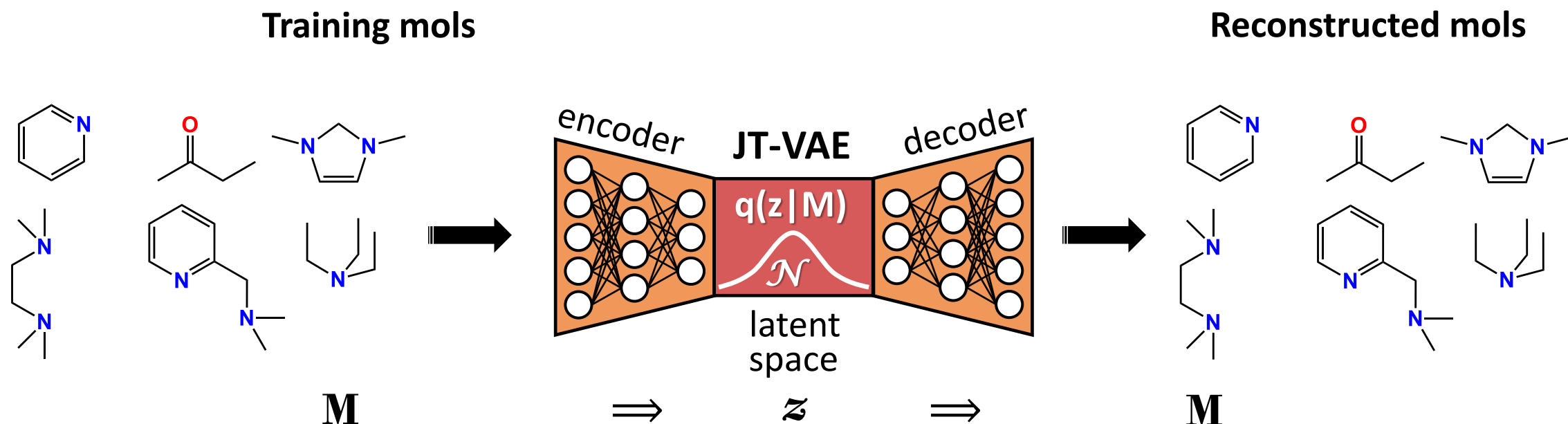


However: Novelty limited to combination of known ligands

Deep generative AI: Junction tree variational autoencoder (JT-VAE)

Model architecture

$$\mathcal{L} = \mathcal{L}_{\text{rec}} + \mathcal{L}_{\text{KL}}$$



Deep generative AI: Junction tree variational autoencoder (JT-VAE)

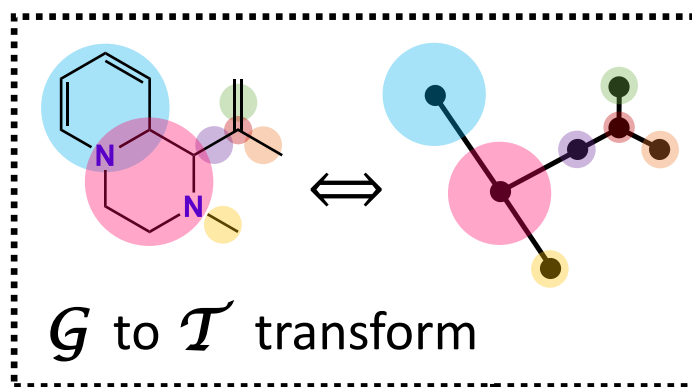
Model architecture

$\mathcal{G}$ = Graph

$\mathcal{T}$ = Tree $\equiv$ Coarse-grained $\mathcal{G}$

Easier to learn $\mathcal{T}$ generation than $\mathcal{G}$

But $\mathcal{T} \rightarrow \mathcal{G}$ should be learned too

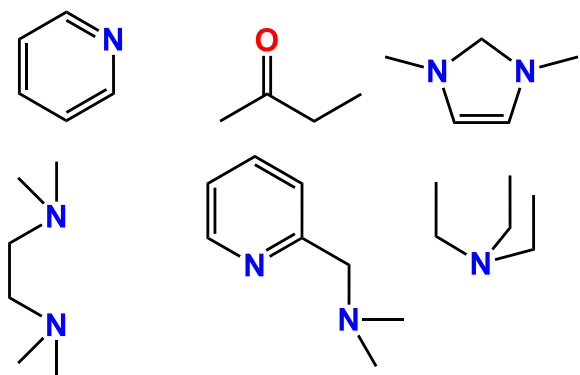


Ring clusters:  

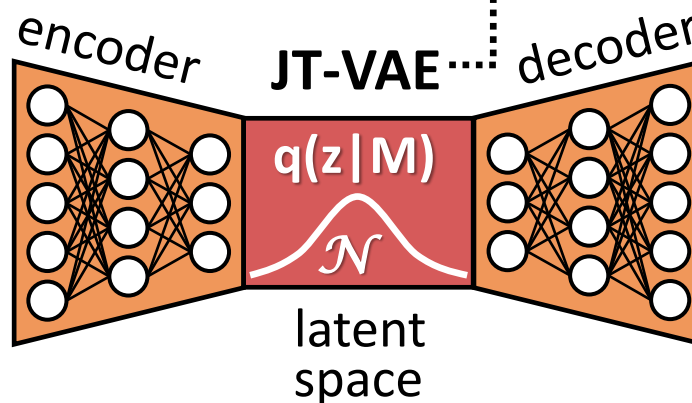
Bond clusters:    

Atom clusters: 

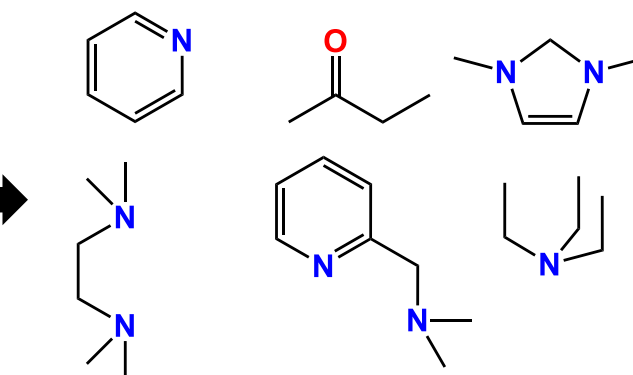
Training mols



$\mathbf{M}$



Reconstructed mols



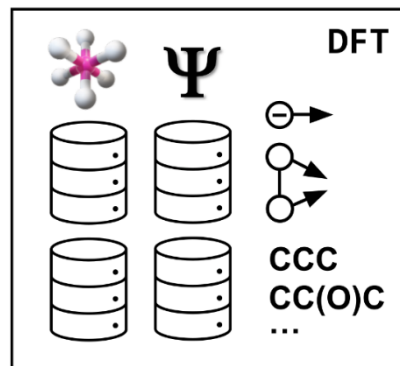
$\mathbf{M}$

JT-VAE with transition metal ligands and complexes?

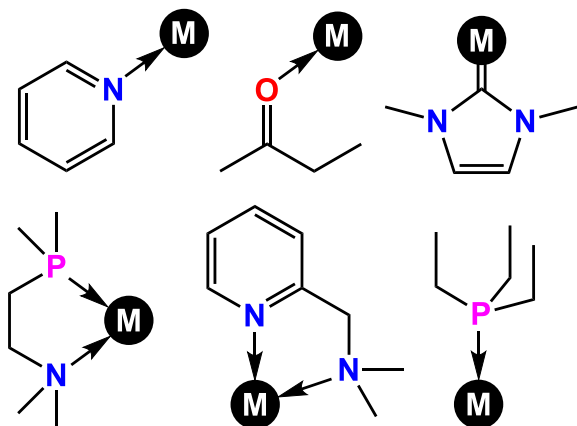
Model architecture for a generator of metal ligands

tmQMg-L

Library (35k ligands)

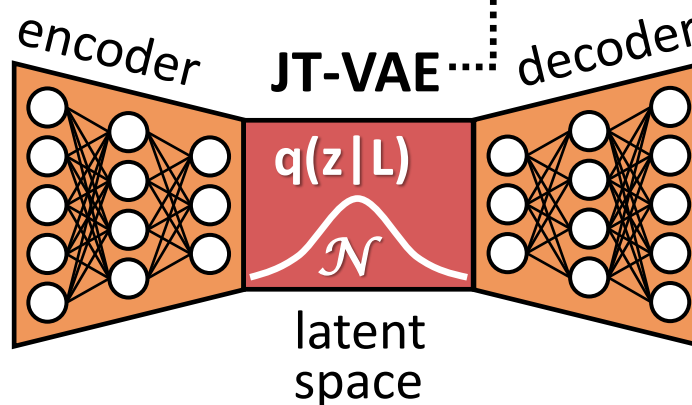


Training ligands



L

TMC $\mathcal{G}$?

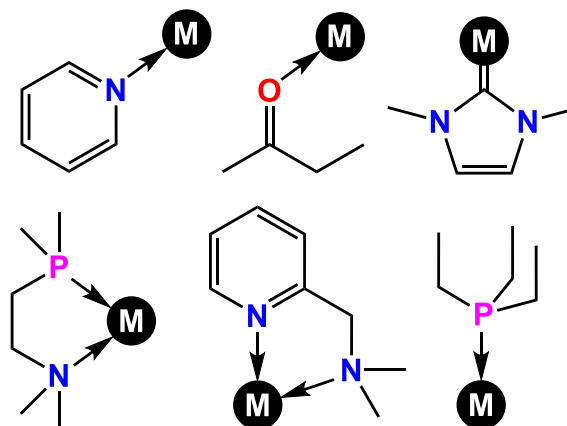


$\Rightarrow$

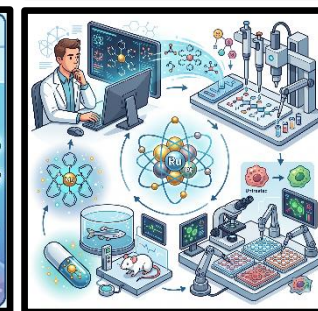
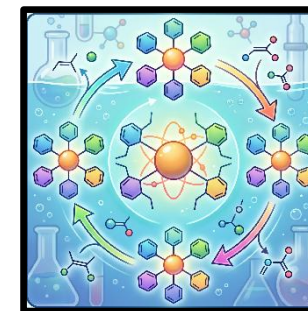
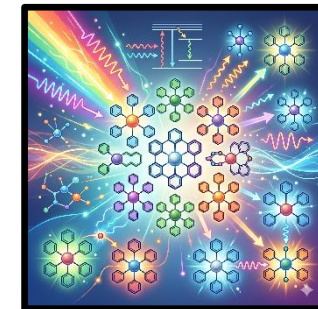
$\mathcal{Z}$

$\Rightarrow$

Reconstructed ligands



L

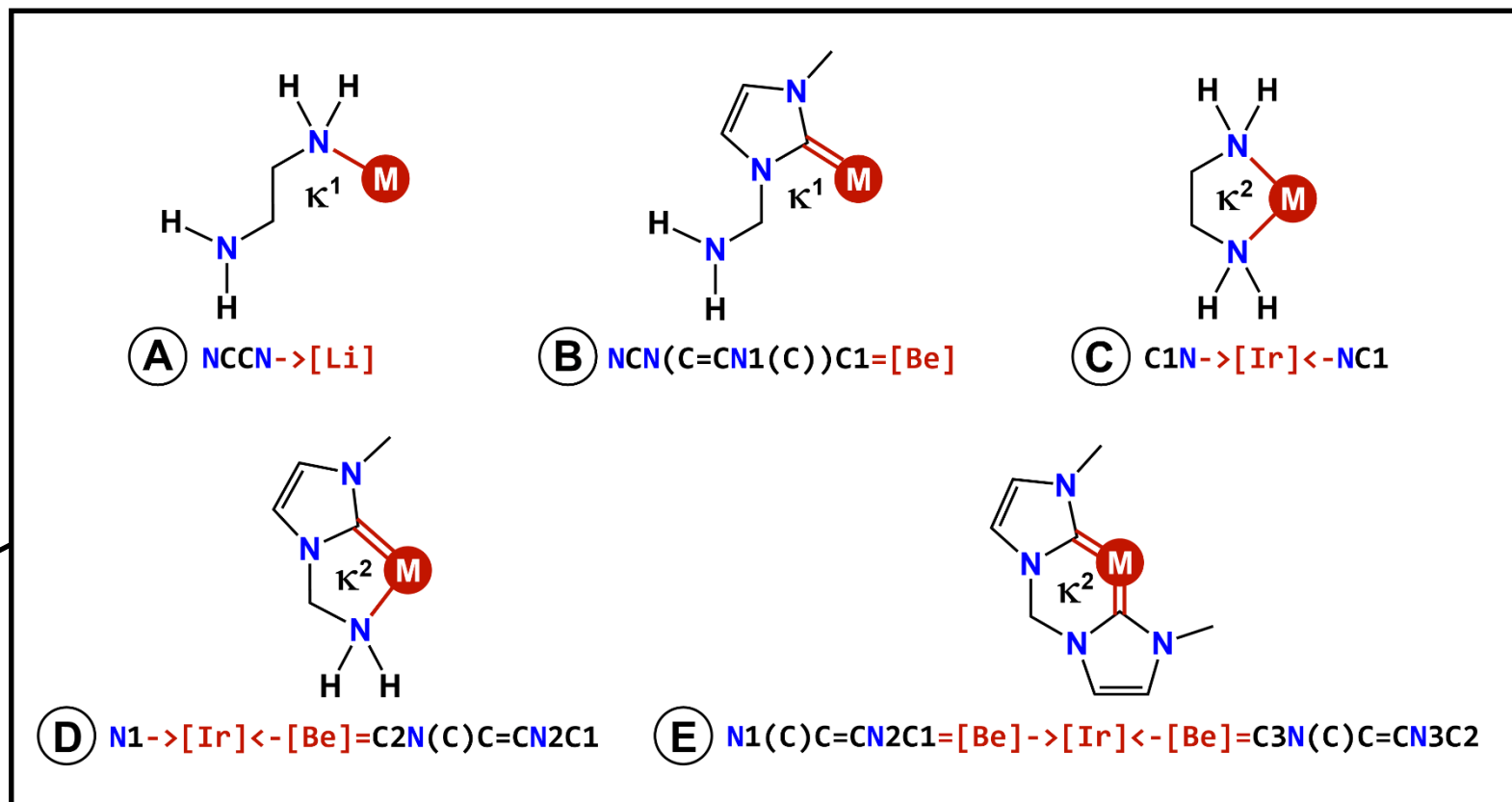
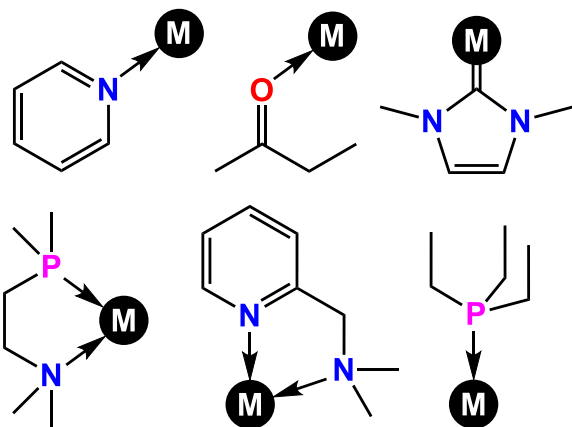
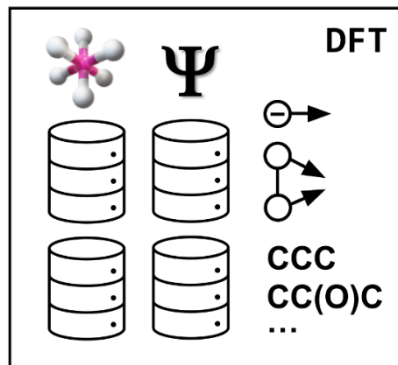


JT-VAE with transition metal ligands and complexes?

SMILES-based string encoding of metal anchor sites and coordination modes

tmQMg-L

Library (35k ligands)

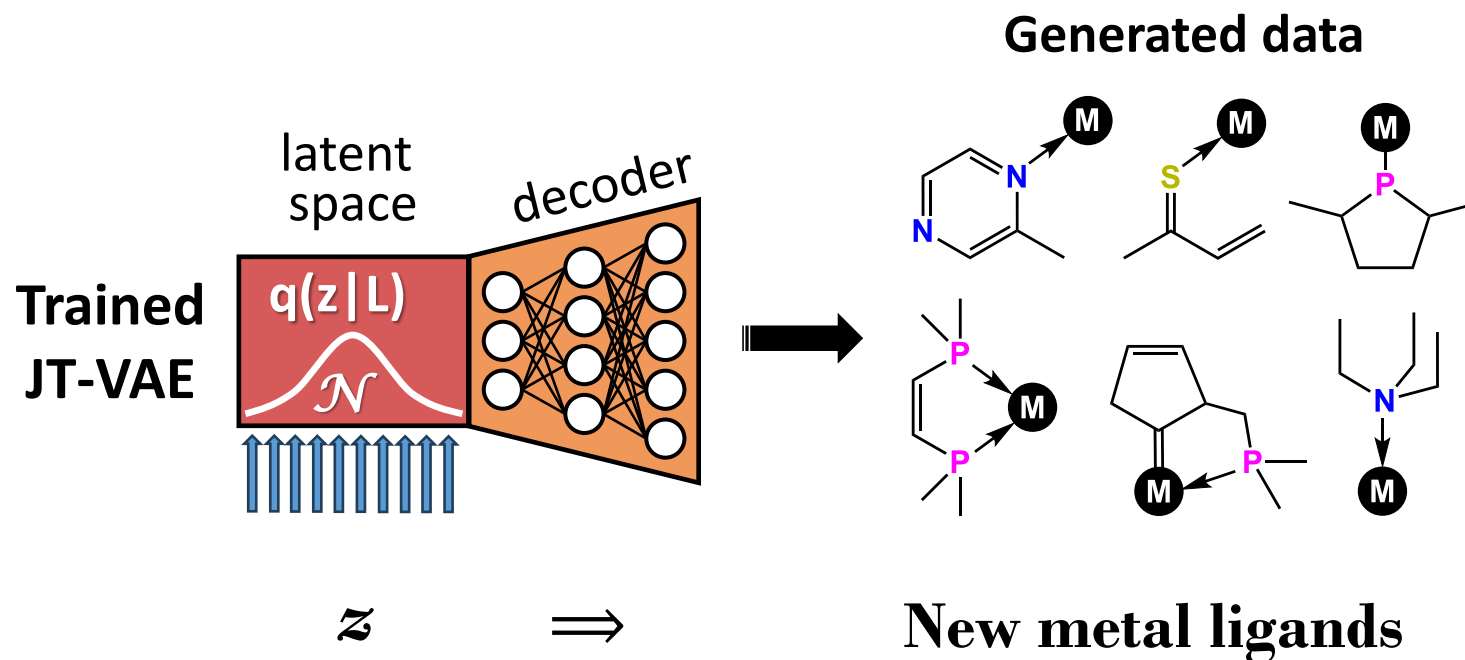


[Li], [Be], [Ir] ↔ Any TM | Metallacycles → Haptic Ls

JT-VAE: Making new ligands

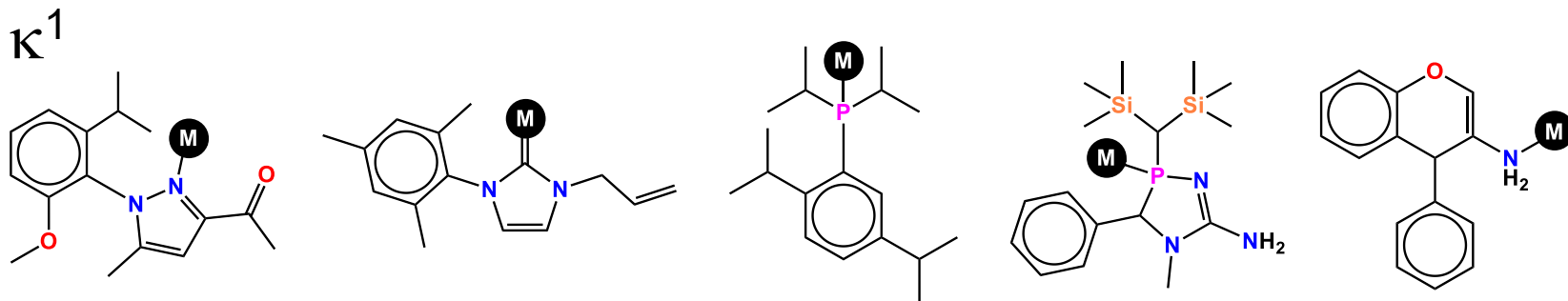
Sampling the latent space and decoding:

Unconditional generation



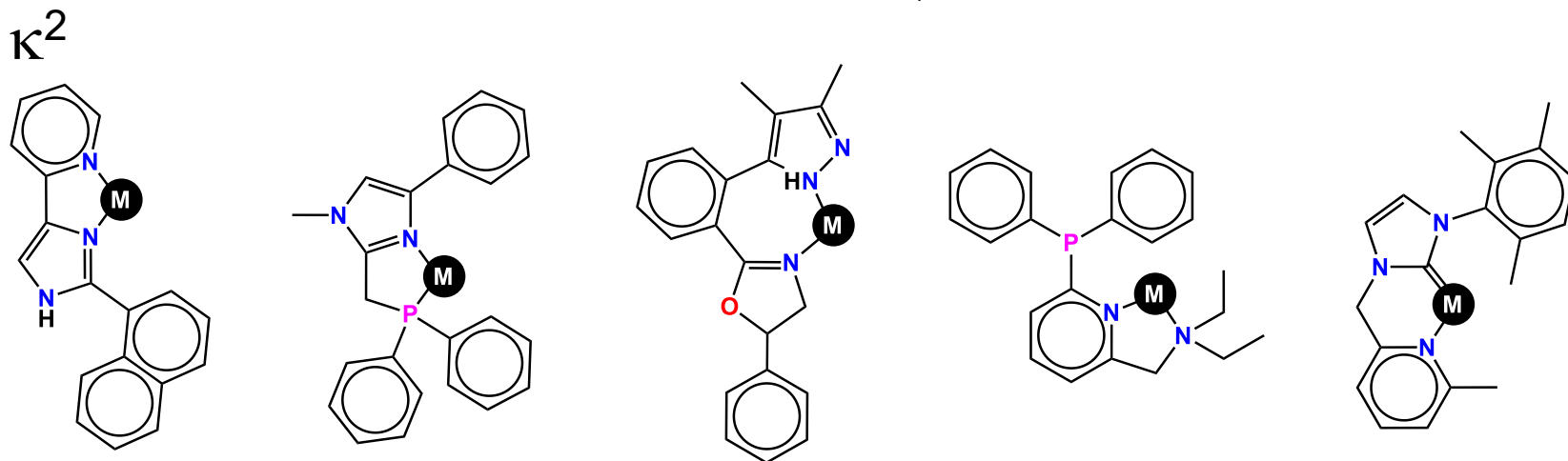
JT-VAE: Making new ligands

Unconditional ligand generation: Validity, novelty, uniqueness → Augmentation



Metrics for whole κ^1 set:

99.6% valid, 69.4% unique
95.5% novel, 7x augmentation
(10k training → 70 k generated)



Metrics for whole κ^2 set:

96.3% valid, 58.7% unique
92.7% novel, 6x augmentation
(10k training → 60 k generated)

Novel $\neq$ Training $\mathcal{D}$ (CSD)

High novelty, high validity, and 6 to 7x data augmentation factors

Ligand synthesizability intrinsically learned from the CSD database

JT-VAE: Inverse design by dual-objective optimization

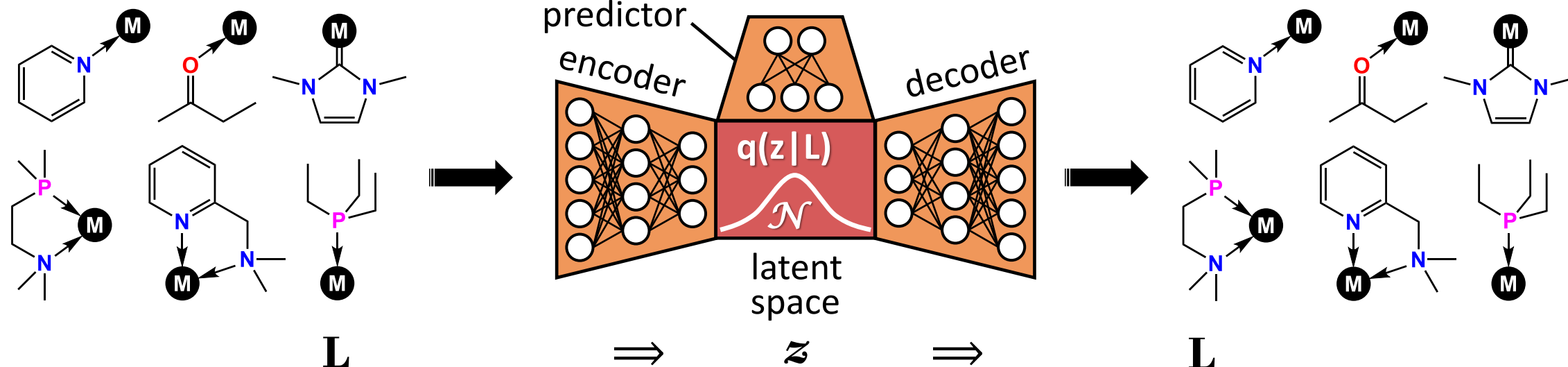
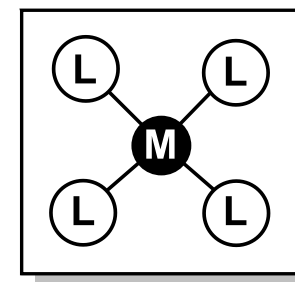
Conditional ligand generation: By two target properties of the homoleptic complex

$$\mathcal{L} =$$

$$\mathcal{L}_{\text{rec}} + \mathcal{L}_{\text{KL}} + \mathcal{L}_{\text{pred}}$$

E.g., ε and $q(\text{M})$

$$\mathbf{Y} = (y_1, y_2)$$

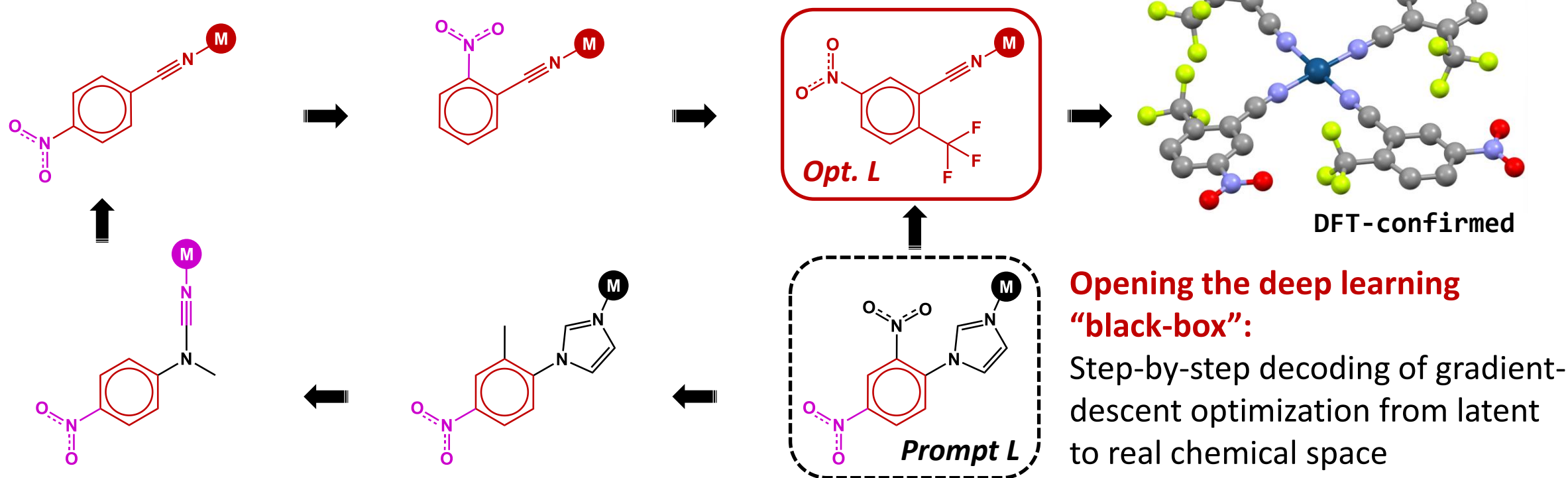


JT-VAE: Inverse design by dual-objective optimization

Conditional ligand generation: By two target properties of homoleptic complex

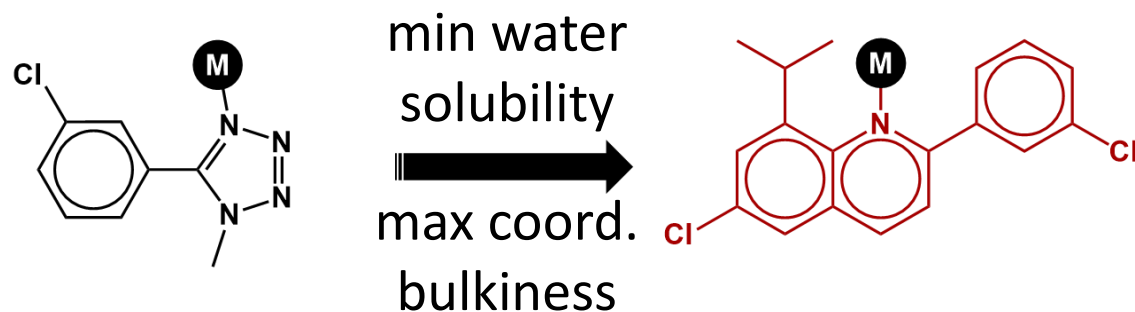
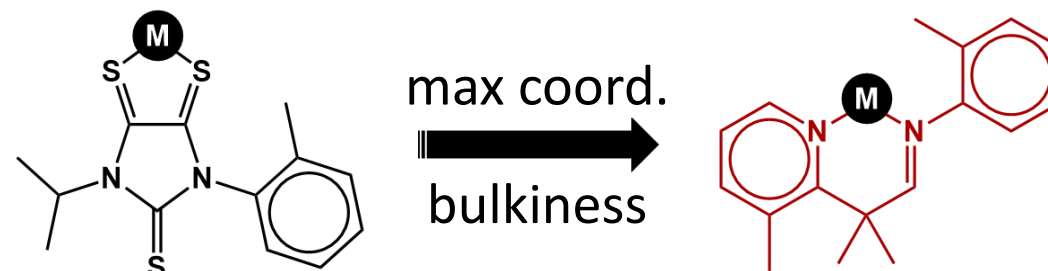
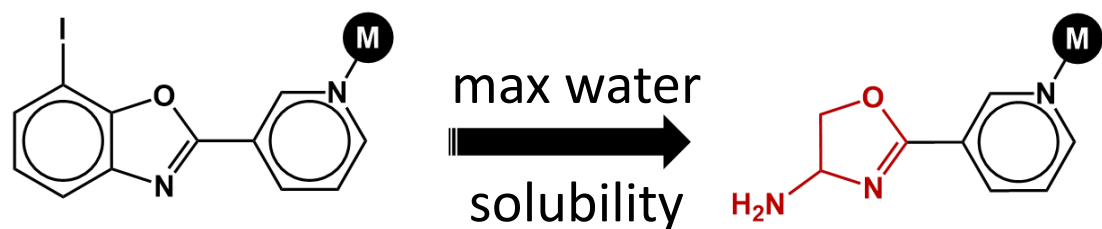
Input: From imine L prompt, make another L maximizing HOMO-LUMO gap & metal center charge for homoleptic square planar Ir(I) TMC

Output: Cyanide L satisfying the prompted condition



JT-VAE: Inverse design by dual-objective optimization

Conditional ligand generation: By two target properties of the ligand itself



However: Upon verification, conditional optimization success rates are rather low; ~50% for TMCs, ~70% for ligands, in dual-objective tasks

Optimality	PL-MOGA ☒	JT-VAE ☐
Novelty	PL-MOGA ☐	JT-VAE ☒

Making the novel exceptional with evolutionary machine learning

Embed DL into GA:

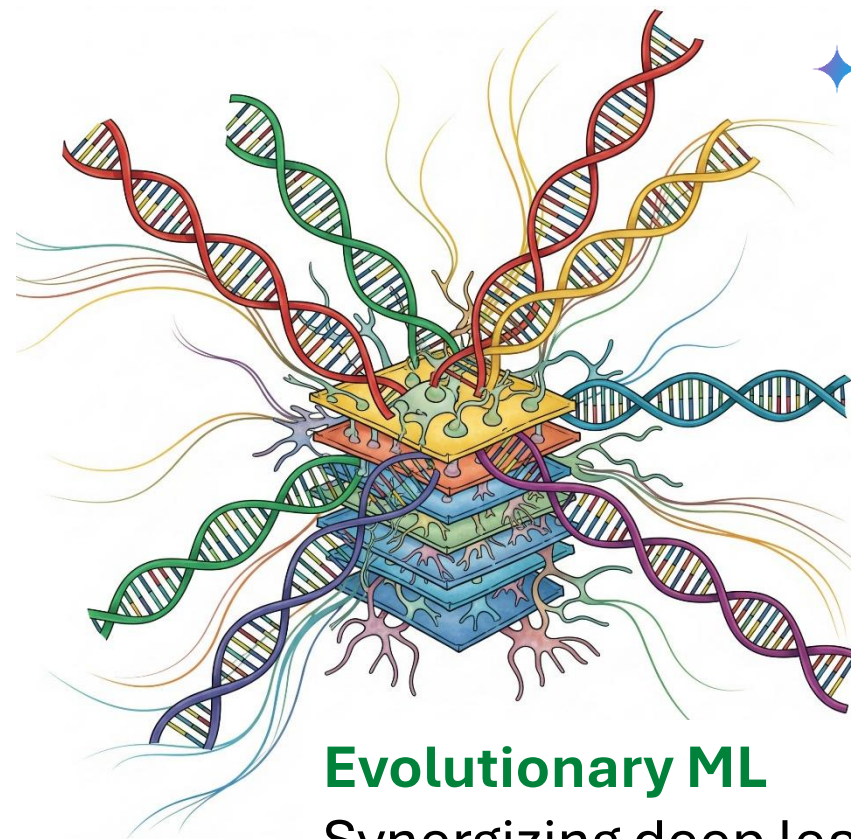
- ⦿ Accelerate the computation of the fitness with a neural network surrogate model
- ⦿ Decide paternity with a neural network classifier

Embed GA into DL:

- ⦿ Do genetic operations in latent space

Couple GA to DL:

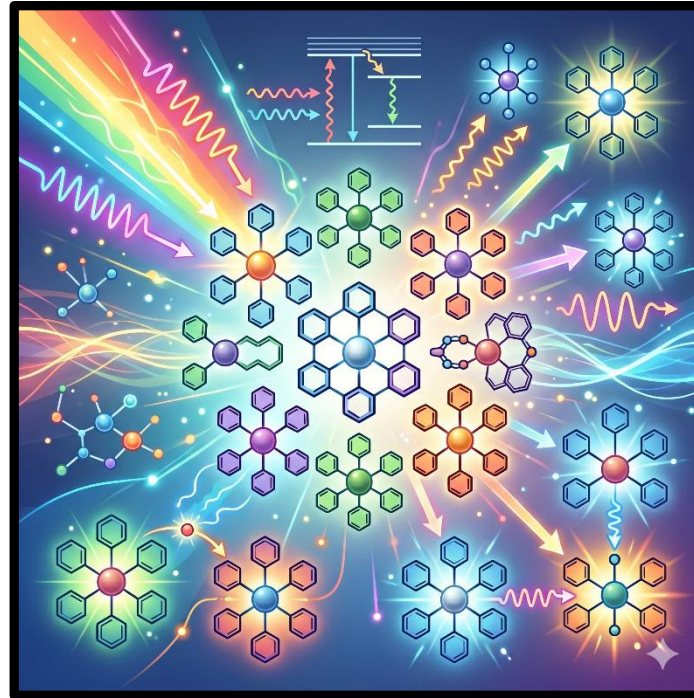
- ⦿ Do genetic operations with molecular fragments generated by a deep generative AI model



Evolutionary ML

Synergizing deep learning with genetic algorithms

Why are we interested in the chemistry of the transition metals?

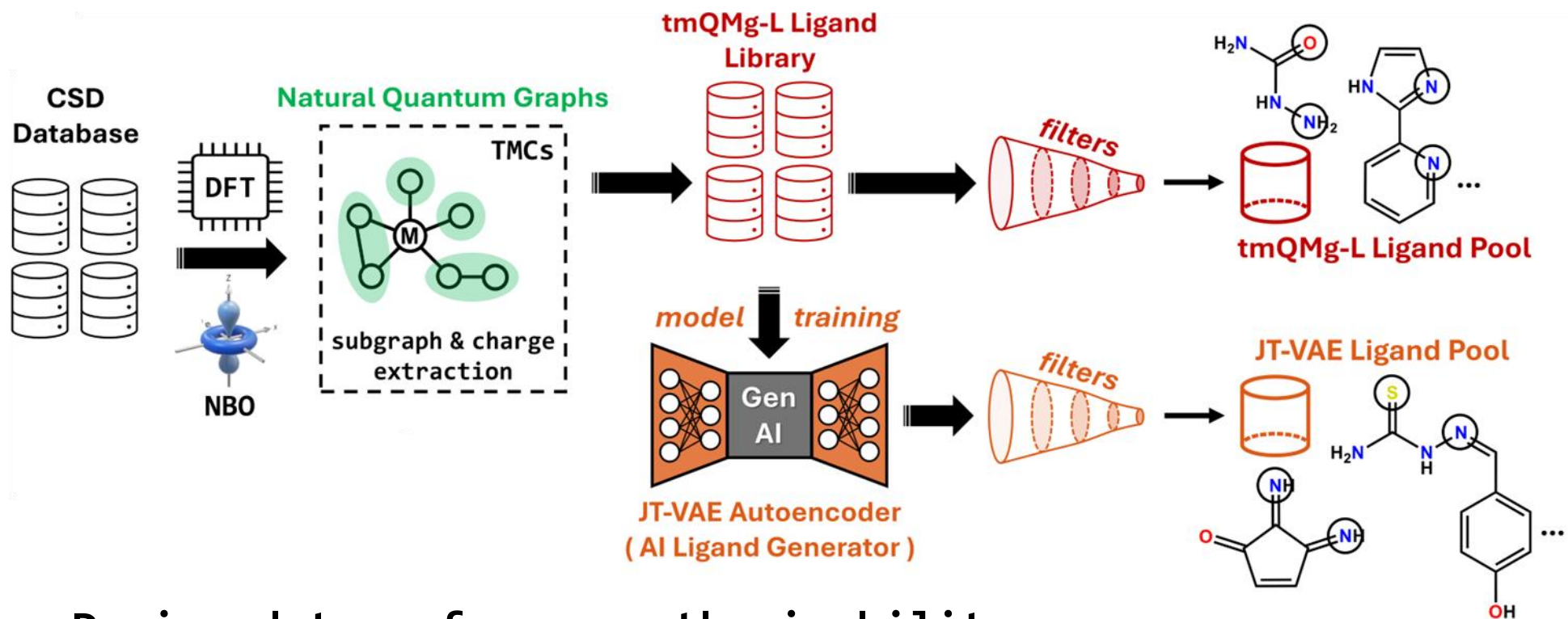


MOLECULAR PHOTONICS

*Intense, wide-spectrum light
absorption, and water-soluble
ruthenium chromophores*

Evolutionary machine learning: Defining ligand pools for the GA

Two ligand pools: tmQMg-L (CSD-known) and JT-VAE (AI-made)



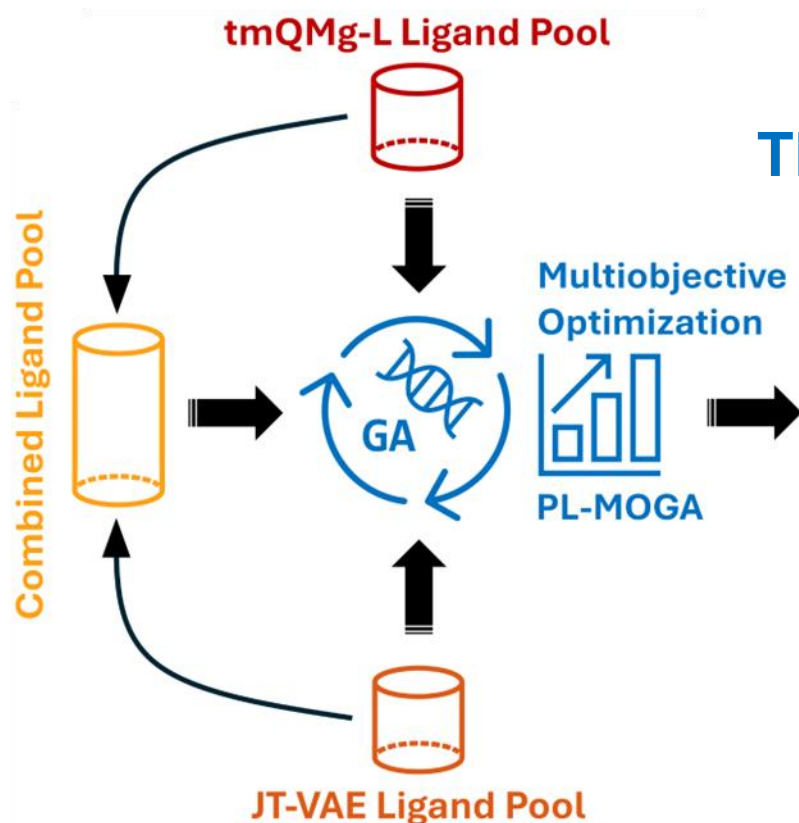
Filters: Designed to enforce synthesizability:

tmQMg-L: Rank by occurrence in Ru TMCs in CSD < Top 75

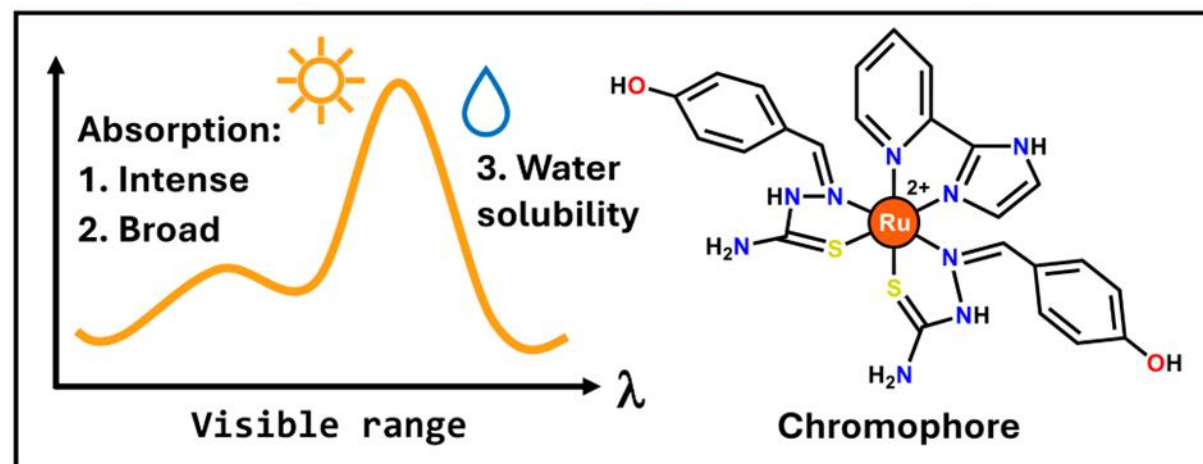
JT-VAE: Rank by synthesizability (SA score) < Top 75

Evolutionary machine learning: Defining 3-objective optimization

A third pool combining tmQMg-L & JT-VAE pools is added for diversity (150 Ls)



Three-Objective Optimization of Chromophores

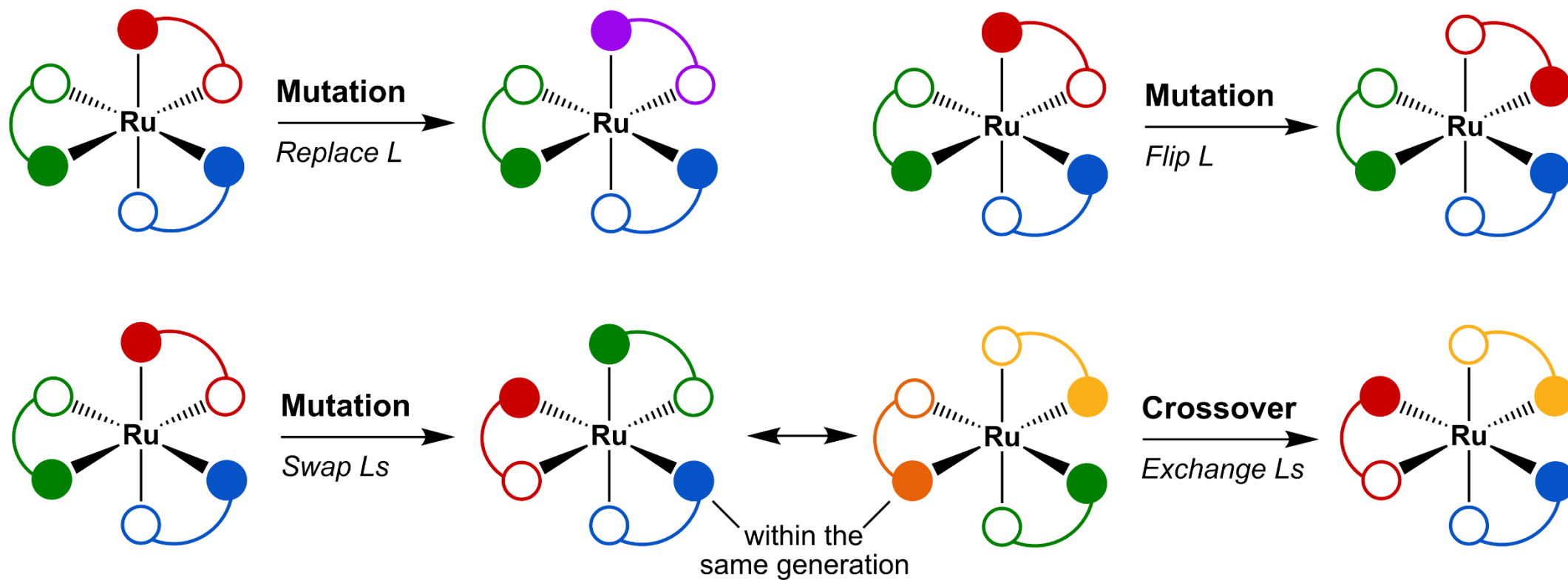


TD-DFT PBE0/LANL2TZ(f),6-31G* evolution
 ω B97xd/def2TZVP hits refinement

Chemical space explored: **3.3M $[\text{Ru}(\text{L})_3]^{2+}$ potential chromophores**

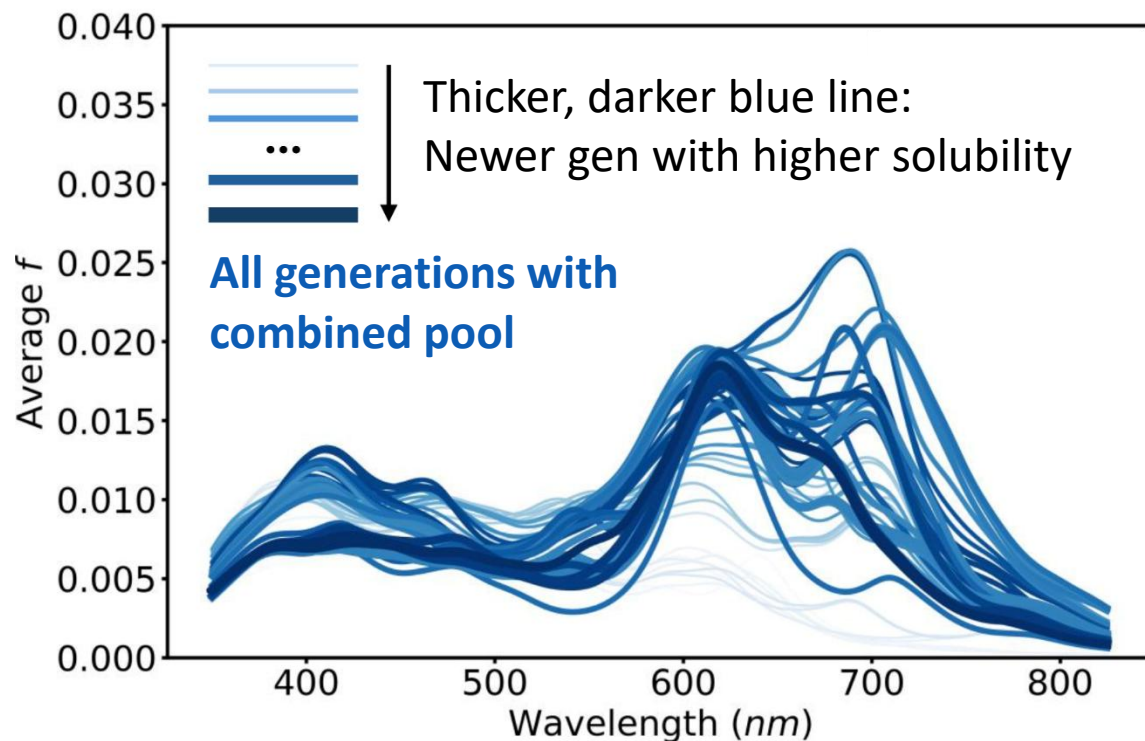
Generating the chromophores: Genetic operations

Besides evolution, these operations also explore the cis/trans isomerization space of the octahedral coordination geometry, including fac/mer isomers



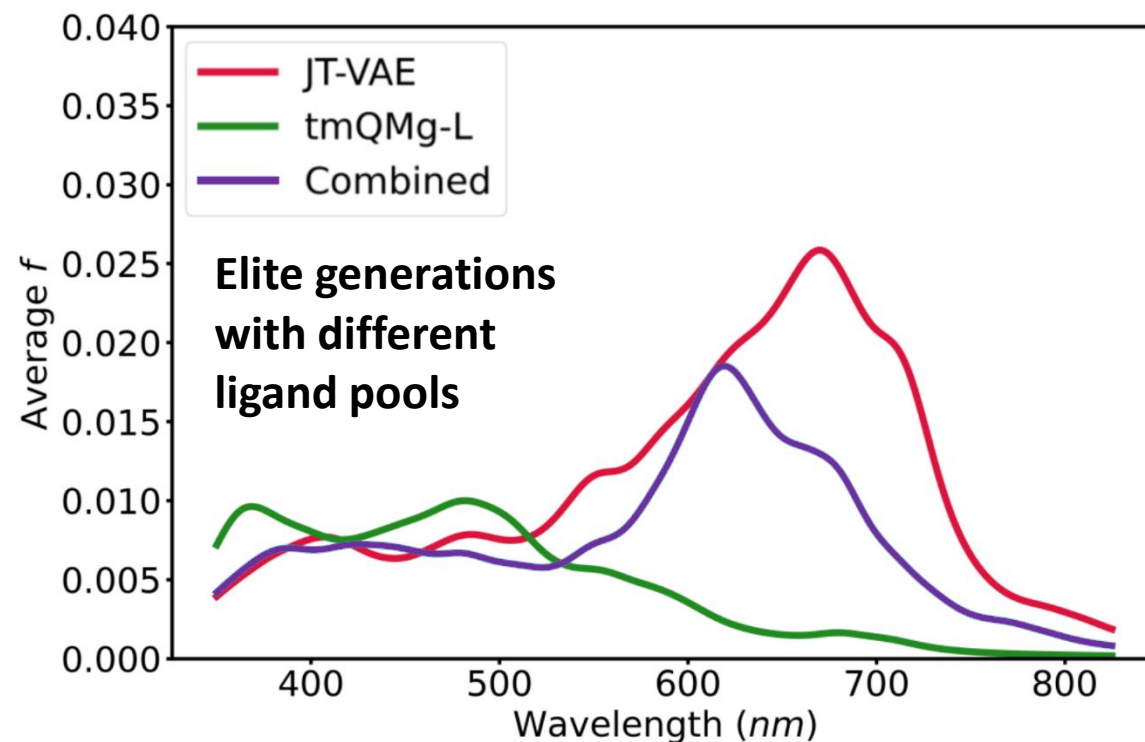
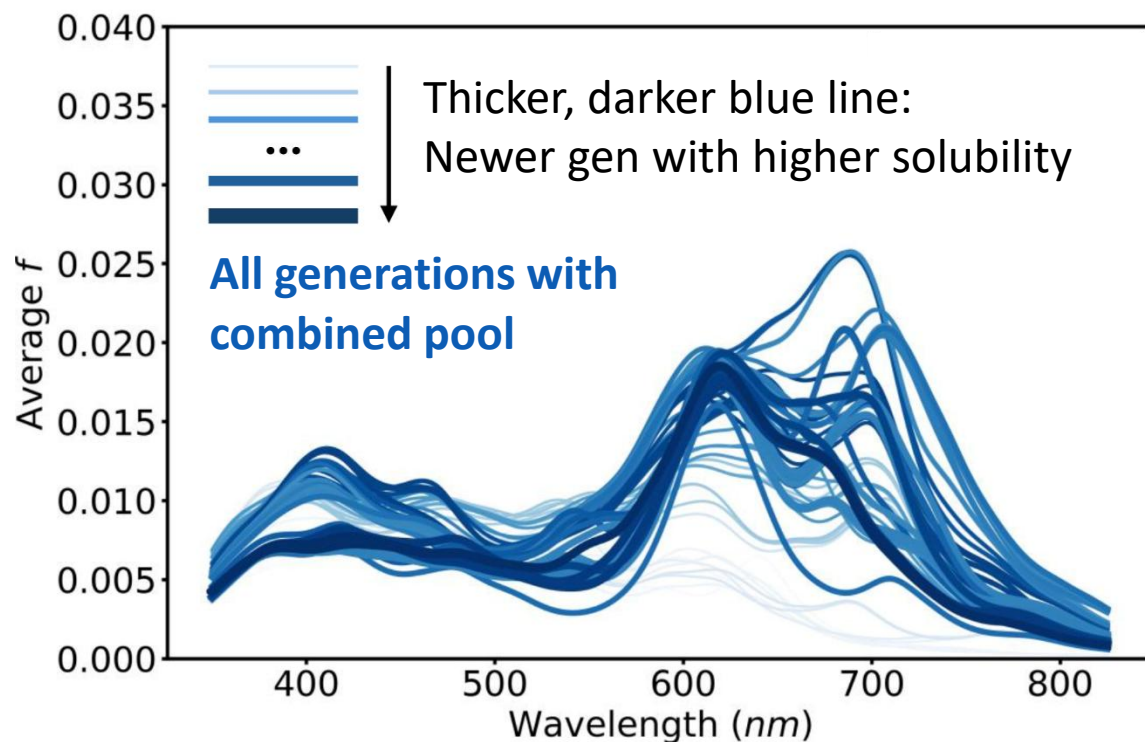
Generating the chromophores: Spectra evolution

Changing shape – One can select generations prior to elite based on this criteria



Generating the chromophores: Spectra evolution

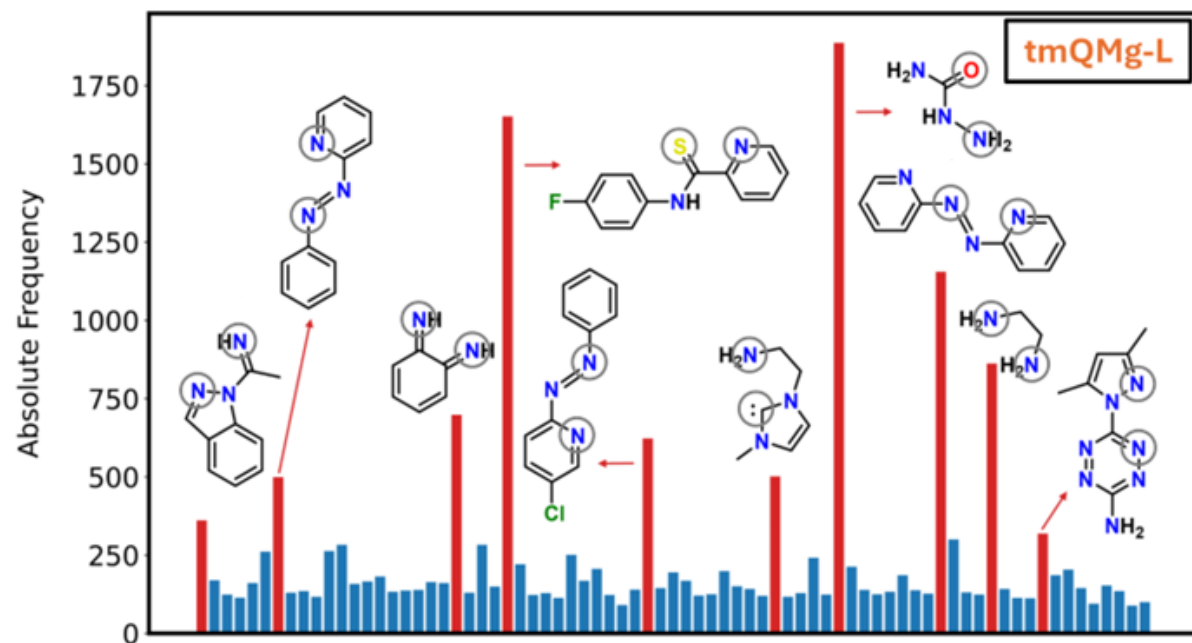
Changing shape – One can select generations prior to elite based on this criteria



tmQMg-L absorbs in near-UV | JT-VAE & Combined both in near-UV and near-IR

Evolutionary machine learning: Opening the “black-box”

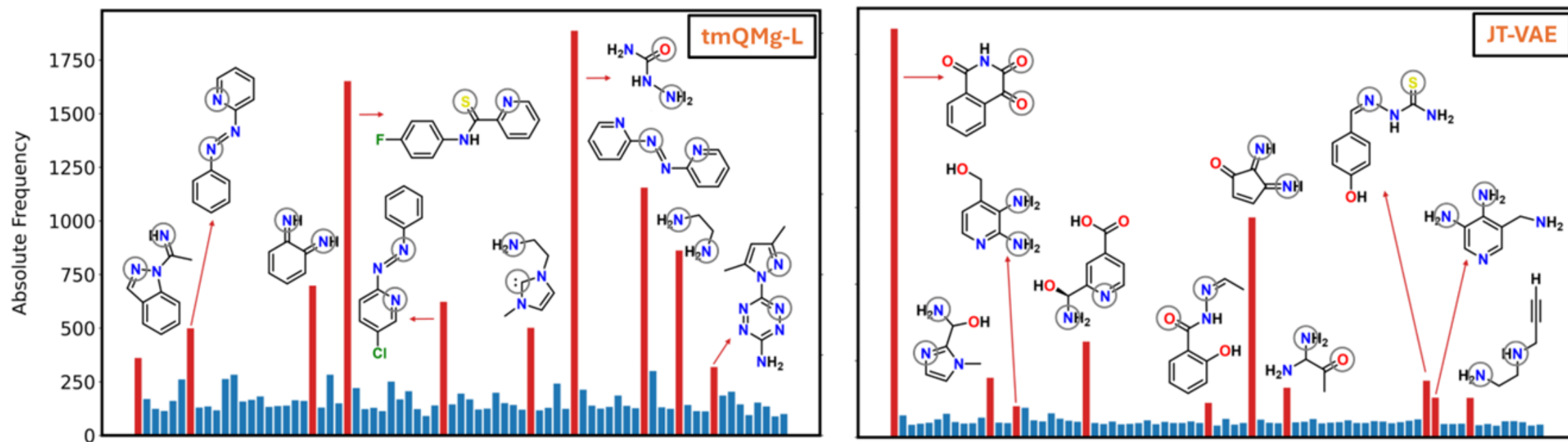
Interpreting the evolution of the hits by analyzing the ligand use frequencies



tmQMg-L: π -conjugated C=N, N=N $\Leftarrow$ Near-UV absorption

Evolutionary machine learning: Opening the “black-box”

Interpreting the evolution of the hits by analyzing the ligand use frequencies



tmQMg-L: π -conjugated C=N, N=N $\Leftarrow$ **Near-UV absorption**

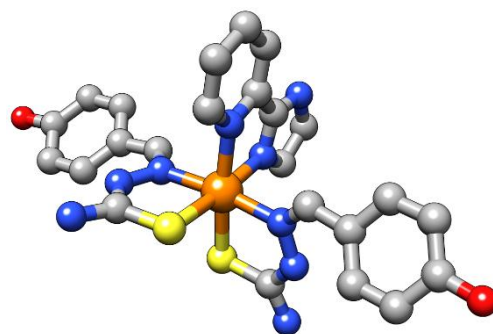
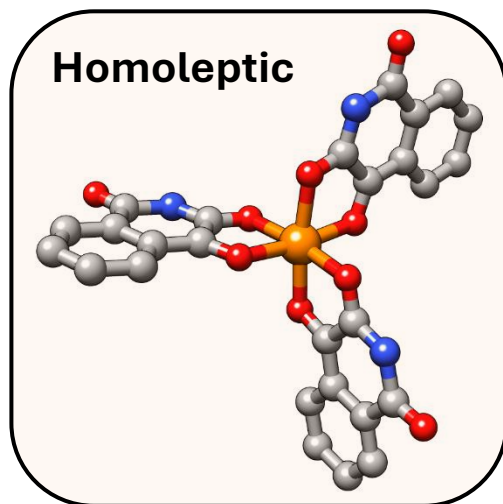
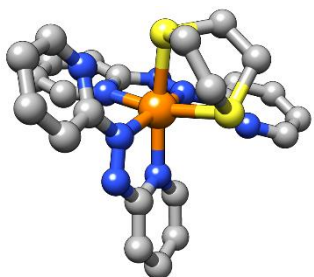
JT-VAE: Adds π -conjugated C=O & asymmetry $\Leftarrow$ **Near-IR absorption, broadness**

JT-VAE: Less rings, more X–H bonds, including O–H $\Leftarrow$ **Increased water solubility**

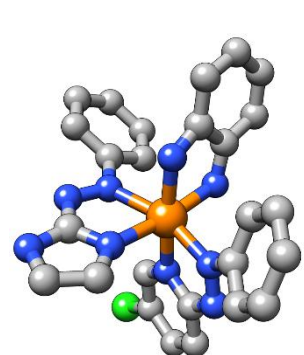
Evolutionary machine learning: Top chromophores for now



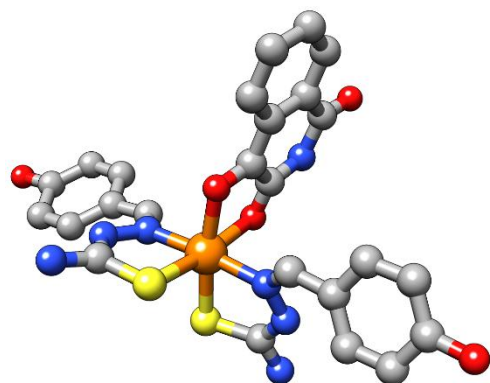
Selected hits:



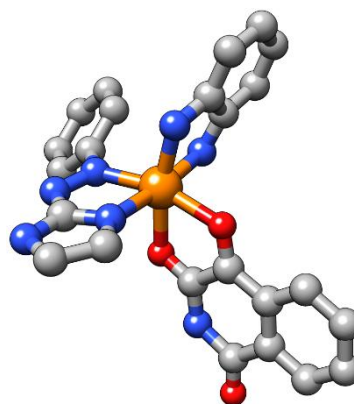
Now trying it in the lab



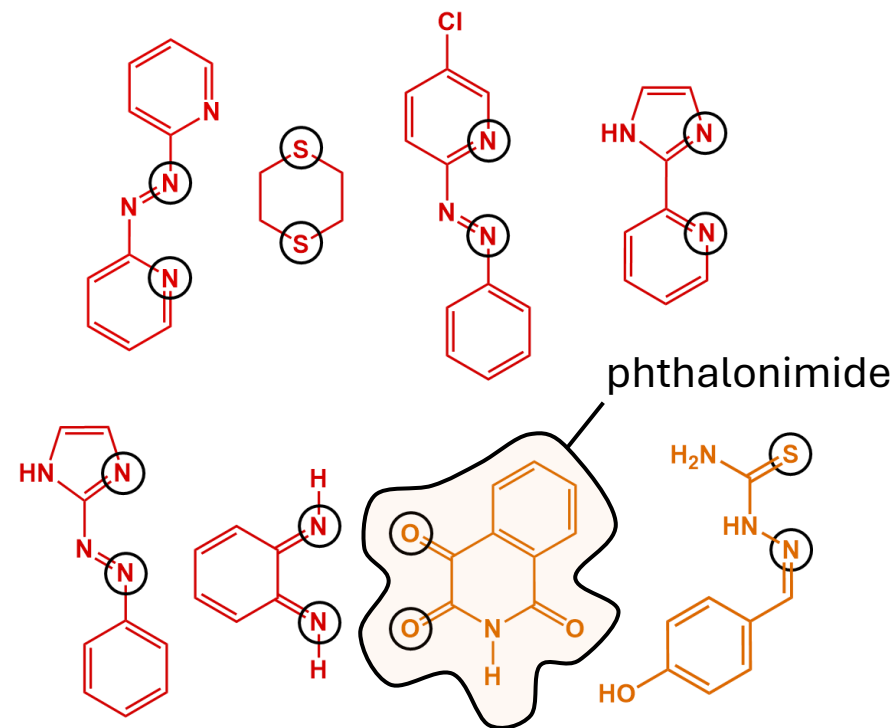
tmQMg-L
(CSD)



JT-VAE



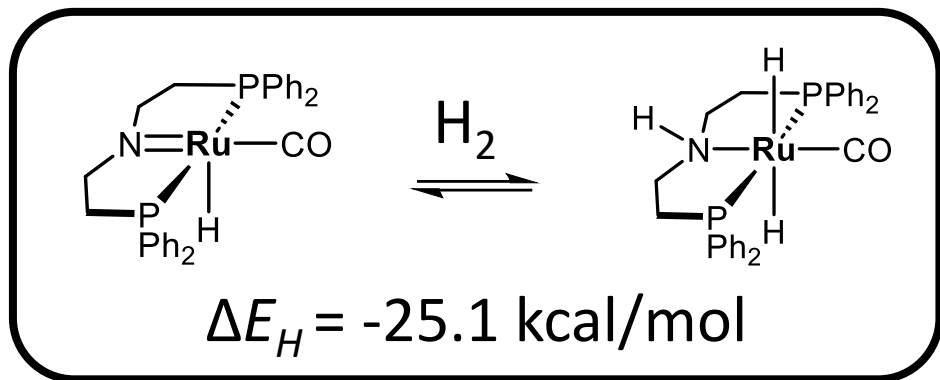
Combined



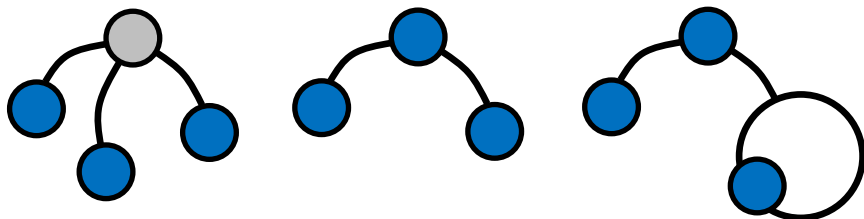
Ligands

(Sustainable CO₂ reduction catalysis in Ru(MACHO) space)

Hydrogenation

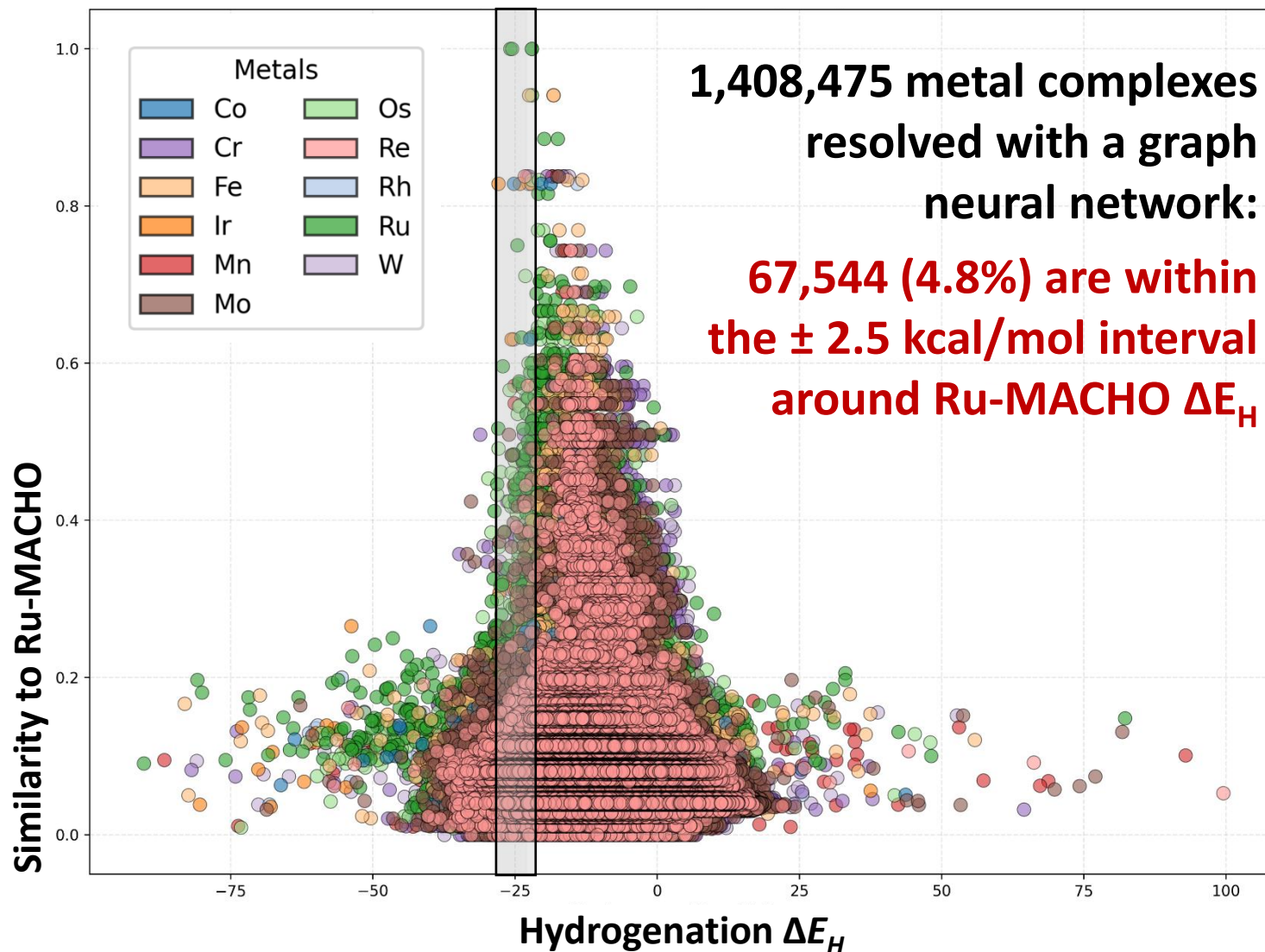


P-free ligands for abundant TMs:



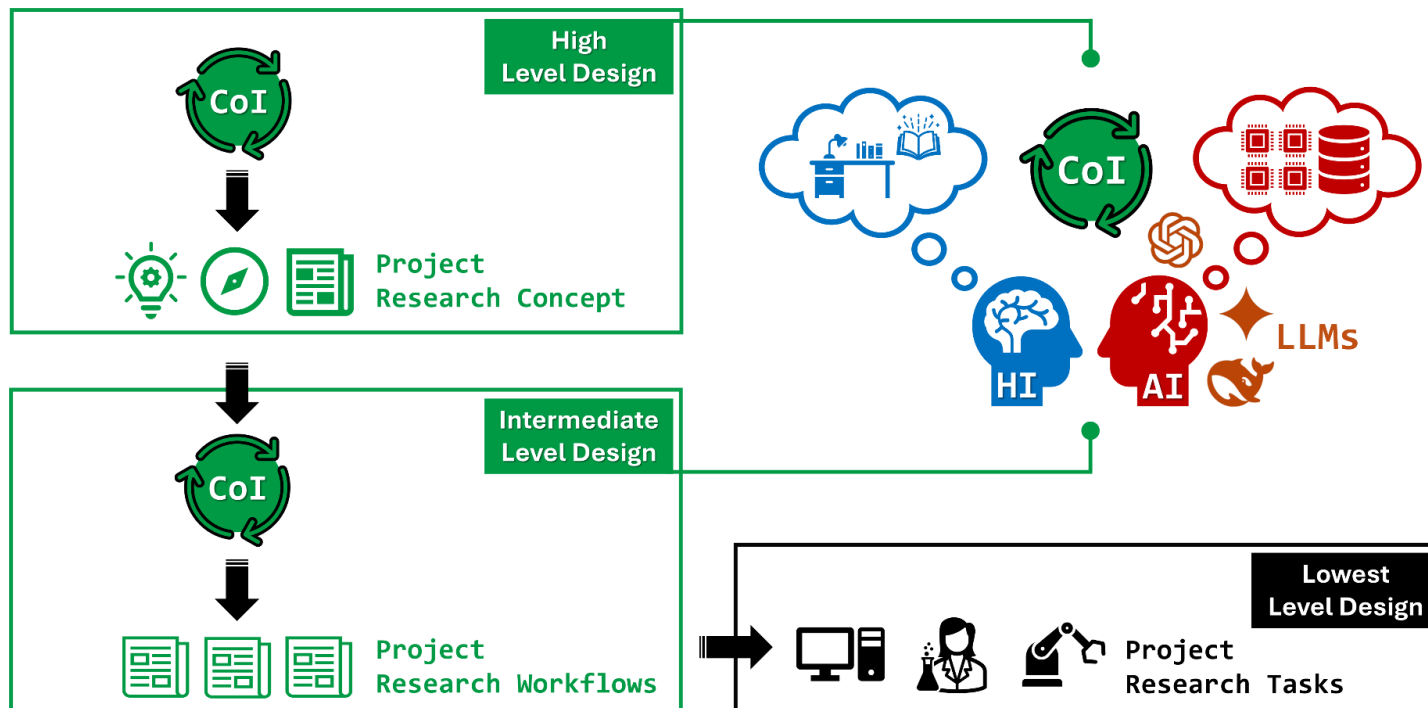
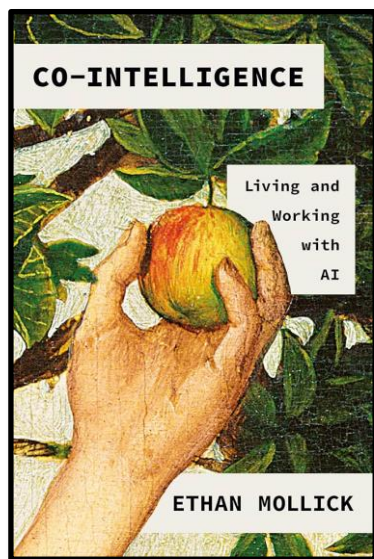
Lucía Morán-González, Ainara Nova (UiO)
Matthias Beller (LIKAT)

$\pm 2.5 \text{ kcal/mol}$ interval: $\Delta E_H = -27.6$ to -22.6



**FINALLY, RECOMMENDED READINGS
THAT COULD BE USEFUL FOR
YOUR FURTHER TRAINING**

If you are interested in using large language models for research



pubs.acs.org/acscatalysis

Viewpoint

Co-intelligent Design of Catalysis Research with Large Language Models: Hype or Reality?

David Balcells*

AI committee: Gemini, DeepSeek, GPT



Cite This: *ACS Catal.* 2025, 15, 16412–16420



Read Online

If you want to know more about AI for homogeneous catalysis

ACS Catalysis > Vol 15/Issue 11 > Article

Open Access

Cite Share Jump to Expand

PERSPECTIVE | May 14, 2025

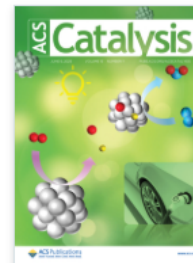
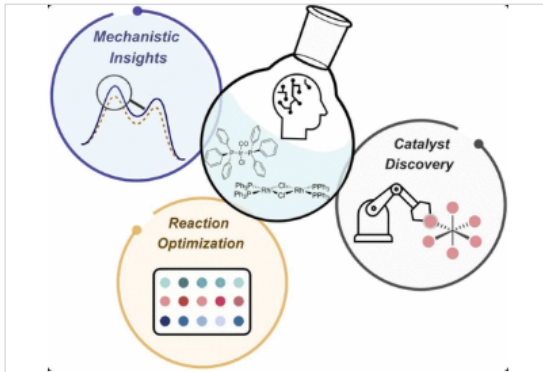
AI Approaches to Homogeneous Catalysis with Transition Metal Complexes

Lucía Morán-González, Arron L. Burnage, Ainara Nova, and David Balcells*

Open PDF

Abstract

Artificial intelligence (AI) is transforming research in chemistry, including homogeneous catalysis with transition metals. Over the past 15 years, the number of publications combining AI with catalysis has increased exponentially, reflecting the interest and strength of this strategy in the field. Since this is a broad emerging discipline, it is essential to establish guidelines that clarify the diverse approaches already available. The complexity of the tasks that can be carried out with AI tools is directly linked to the nature of their components, including datasets, representations, algorithms, and high-throughput experimental and computational facilities. In parallel to the evolution of these tools, applications to catalysis have also advanced. Initially, models were developed to predict key aspects of the reaction mechanism, aiming at screening catalyst candidates. Subsequent studies have incorporated experimental data to optimize reaction conditions and yields. More recently, generative AI based on deep learning methods has enabled the inverse design of novel catalysts with predefined target properties. While most studies rely on computational data, recent advancements have improved the acquisition of experimental data, enabling AI-driven automated workflows. This Perspective gives a critical overview on selected studies that reflect the state of the art in the application of AI to homogeneous metal-catalyzed reactions, also highlighting future opportunities and challenges.



ACS Catalysis

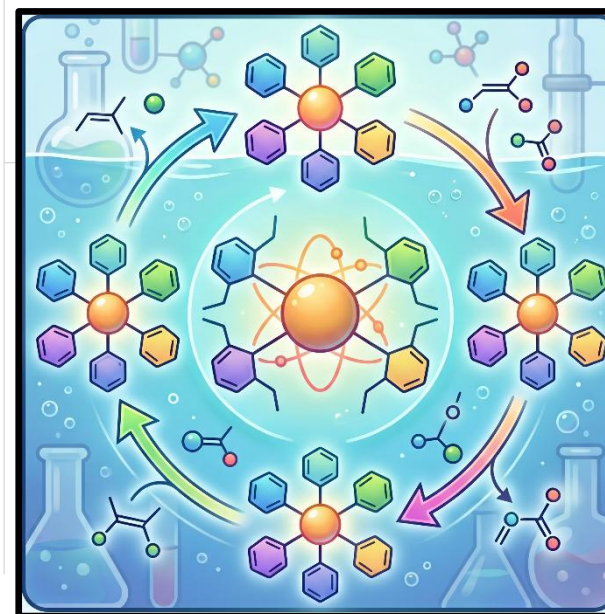
Cite this: *ACS Catal.* 2025, 15, 11, 9089–9105

<https://doi.org/10.1021/acscatal.5c01202>

Published May 14, 2025

Copyright © 2025 The Authors. Published by American Chemical Society. This publication is licensed under [CC-BY 4.0](#).

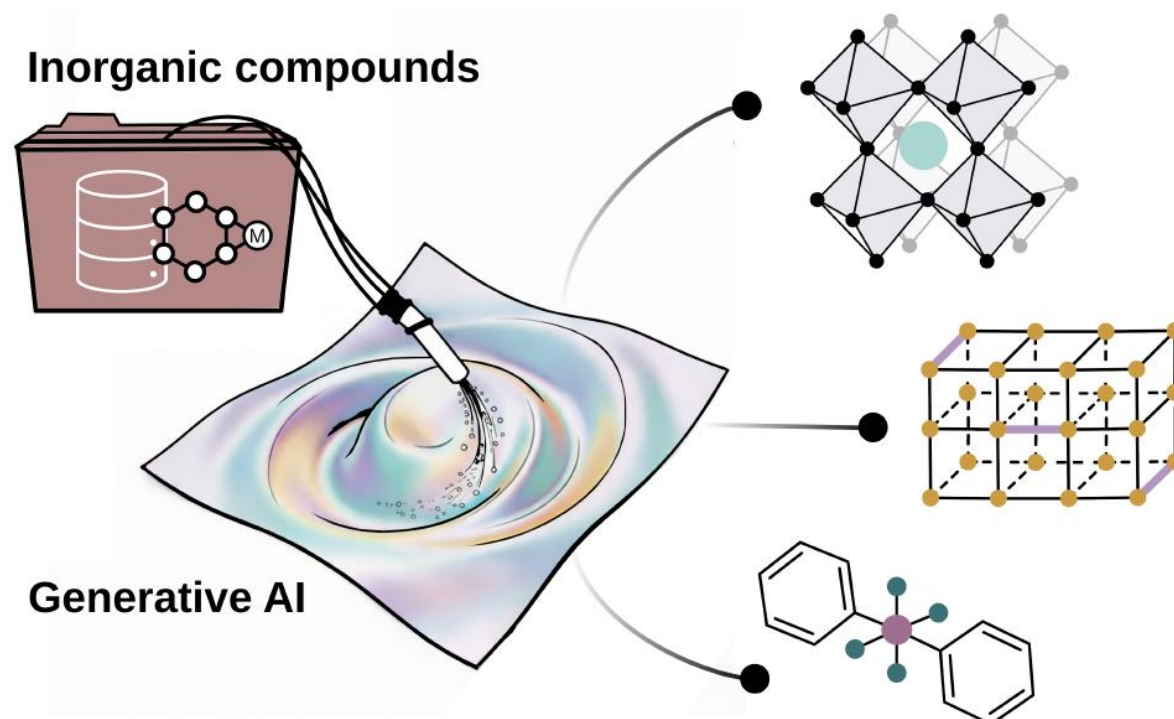
Get e-Alerts



If you want to know more about generative AI for metal complexes and, generally, inorganic compounds (MOFs, perovskites, ...)

Inverse Design of Inorganic Compounds with Generative AI

Hannes Kneiding¹, Lucía Morán-González¹, Nishamol Kuriakose¹, Ainara Nova¹, and David Balcells^{1,*}



Generative AI for transition metal chemistry (making the novel, exceptional?)



David Balcells



THANKS FOR YOUR KIND ATTENTION !!

